College Majors and Skill Mismatch: From Micro Wage Losses to Macro Output Losses*

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Abstract

This paper studies the aggregate and distributional consequences of college major–occupation mismatch. While most existing research focuses on individual-level wage losses, we develop a framework to quantify the aggregate inefficiencies from mismatch. We define mismatch as the deviation from the efficient allocation of college majors to occupations, given observed wage returns. We estimate these returns using a Roy model applied to unique Australian administrative tax panel data linking university degrees to the longitudinal earnings and employment histories. Consistent with prior work, we find that mismatch leads to substantial individual wage losses—averaging 27 percent with fixed skill prices. However, the corresponding aggregate output losses are more limited, about 10 percent when skill prices adjust. Importantly, mismatch declines over the life cycle and largely vanishes by age 30 with variable prices. The findings underscore how micro-level misallocation in human capital may not always aggregate to output inefficiencies.

Keywords: Skill Mismatch, College Major, Occupation, Roy Model, Misallocation, Administrative Tax Record

JEL Classification: E24, J24, J31.

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1 Introduction

Tertiary education plays a central role in a country's economic development by equipping individuals with skills that are expected to meet labor market needs. In most systems, students freely choose their fields of study, often guided by beliefs about future labor demand. However, due to rapid technological change and evolving industry needs, these expectations can be misaligned with actual labor market outcomes. As a result, many graduates find themselves in occupations that do not fully utilize their acquired skills—a phenomenon known as skill mismatch.

Skill mismatch is increasingly a concern for economists and policymakers.[1] At the individual level, mismatch can reduce wages and hinder career progression; at the aggregate level, it may distort labor allocation, exacerbate inequality, and generate inefficiencies in production. Existing empirical research has focused largely on micro-level measures of mismatch based on individual wage estimates, typically estimating wage penalties for individuals working outside their field of training. While such estimates are informative about the consequences of mismatch for individual workers, they need not capture its aggregate costs. Policies aimed at reducing mismatch can induce labor reallocation, changing the relative supply of skills and, consequently, their equilibrium returns. These price adjustments change the gains from reallocation implied by individual-level estimates. Thus, assessing the aggregate (macro) consequences of mismatch requires going beyond individual-level (micro) estimates that hold returns fixed and accounting for how returns adjust as workers are reallocated across the labor market.[2]

This paper quantifies college major–occupation mismatch at both the individual and aggregate levels. We first use a Roy model of occupational choice to estimate the returns to each major–occupation pairing while accounting for selection into occupations on observed and unobserved characteristics. We define individual-level, or micro, mismatch as the wage loss from working in one's observed occupation rather than the occupation offering the highest return for one's major, evaluated at prevailing labor market prices.

To assess the aggregate efficiency consequences, we embed these estimates in a model where occupational output is produced using a nested CES technology that aggregates workers by major within occupations. Using regional and temporal variation in labor markets, we estimate the elasticities of substitution of college major by occupation and the elasticity of substitution between occupations for an aggregate production function. We then compute the efficient allocation of workers across occupations, the allocation that maximizes aggregate output and

[1] See, for example, the discussions in OECD (2024).

[2] We refer to mismatch with fixed returns/prices as individual or micro and to mismatch with variable (equilibrium) prices as aggregate or macro throughout the paper.

at which marginal products are equalized within each major, and compare it to the observed allocation to quantify aggregate output losses from mismatch.

We apply this framework to the Australian labor market using the ALife dataset (Carter et al., 2021), a unique administrative panel dataset that allows us to link individuals' tax records with their university degrees. We find substantial individual-level mismatch, with large wage penalties from working outside one's top-return occupation. After accounting for selection, these penalties are large, average 27 percent, and are particularly pronounced for STEM, Management and Commerce, and Arts and Humanities graduates.

However, when labor market prices adjust, the aggregate output losses from mismatch are far smaller, about 10 percent. The gap is largely a price effect: valued at the observed returns, the planner's efficient reallocation yields gains comparable to the individual ones, whereas valued at the returns that prevail after it, the gains are far smaller. As workers move into the occupations where their major earns the most, the returns there fall, and most of the individual gain is competed away. Similar to individual mismatch, most of the aggregate loss is concentrated in a few major groups: Management and Commerce (4.1%), Society, Culture, and Arts (3.2%), and STEM (2.8%). Importantly, mismatch declines significantly over the life cycle as workers switch occupations. Macro mismatch largely disappears by age 30, suggesting that reallocation helps mitigate initial misallocation.

A concern with these life-cycle patterns is that graduates who start in an occupation that does not match their college-major skills may acquire the necessary skills through experience in the occupation itself. Apparent mismatch early in the career may therefore diminish with occupational tenure rather than reflect a permanent misallocation. To address this, we extend the model to incorporate occupation-specific human capital, letting major-occupation returns rise with occupational tenure and letting the planner internalize that reallocated workers arrive at their destination without tenure. Because a switcher loses the occupation-specific human capital accumulated in their current job, the gains from reallocation are smaller and measured mismatch declines both with or without fixed prices. The two main results are nonetheless unchanged: (1) mismatch still falls sharply over the life cycle, and (2) aggregate output losses remain far smaller than the individual wage losses under fixed returns.

As a further robustness check, we examine how the aggregate loss depends on the curvature of the technology. Holding the estimated productivity weights at their baseline values, we vary the within-occupation elasticity of substitution between majors, a margin that models without major-specific inputs do not have, and the between-occupation elasticity over the range of values in the literature. When majors are complements, an occupation's output falls sharply as its mix of majors departs from the efficient one, so the gain from reallocation is larger; when majors are close substitutes, the mix of majors within an occupation matters less for output,

and so does misallocation. Across the grid, the aggregate loss ranges from about 8 to 35 percent. The largest losses arise in the Cobb–Douglas case, where majors are least substitutable, and our baseline lies near the lower end of the range. The loss is much less sensitive to the between-occupation elasticity, on which the literature usually focuses: moving it from the baseline to either end of the range of the literature changes the loss by less than seven percentage points.

Finally, we ask which skills drive mismatch, treating each major as a bundle of O*NET knowledge skills and decomposing the wage losses skill by skill.[3] We find no common skill factor (e.g., mathematics) driving wage losses or mismatch across majors. Instead, the skills that drive mismatch are those that define each field's human capital: engineering and technology skills for STEM graduates, business and management skills for Management and Commerce graduates, and law and public safety skills for Society, Culture, and Arts graduates. This suggests that mismatch arises when workers cannot employ their field-specific human capital which generally also commands the highest return, rather than from a shortfall in some general, transferable skill.

Literature Review

A growing literature measures skill mismatch at a micro-level by comparing workers' qualifications with the requirements of their jobs, often using detailed occupational data from O*NET or similar sources.Early contributions by Pellizzari and Fichen (2013) and Adalet McGowan and Andrews (2015) developed widely used mismatch indices, including those adopted by the OECD. More recent work estimates models of occupational choice using multidimensional skill measures and sorting behavior, often combining longitudinal survey data with occupation-level skill profiles from O*NET (Guvenen et al., 2020; Lise and Postel-Vinay, 2020; Baley et al., 2022).[4,5] Our paper offers an alternative framework that defines mismatch through the observed wage returns to major–occupation pairs, allowing us to estimate the efficiency loss of deviations from the efficient allocation, even in the absence of fine task data. Our approach is conceptually similar to the misallocation literature (Hsieh and Klenow, 2009), as we infer allocative inefficiencies from dispersion in major-specific returns across occupations reflected in wage returns.

[3] O*NET (the Occupational Information Network) is the U.S. Department of Labor's occupational database, which rates each occupation on standardized scales of the knowledge, skills, abilities, and work activities it requires, based on surveys of incumbent workers and occupational analysts.

[4] Instead of O*NET, Bandiera et al. (2024) use microdata from the Programme for the International Assessment of Adult Competencies (PIAAC) to estimate a matching model of workers to jobs.

[5] Other studies define mismatch empirically as gaps between entrants and incumbents (Fredriksson, Hensvik, and Skans, 2018), or between workers and their peers (Groes, Kircher, and Manovskii, 2015). A related macro literature infers mismatch unemployment from the dispersion in vacancy to unemployment ratios across sectors (Şahin, Song, Topa, and Violante, 2014; Patterson, Şahin, Topa, and Violante, 2016).

A second related strand of literature focuses on the estimation of occupational returns using structural models of occupational choice, particularly the Roy model (Roy, 1951). This literature emphasizes the role of self-selection and comparative advantage in shaping observed wage outcomes (French and Taber, 2011), and has developed methods to address high-dimensional selection problems using control function approaches (Lee, 1983; Dahl, 2002). More recently, Eckardt (2026) extends the Roy framework to analyze training–occupation mismatch in the German apprenticeship system. Our paper builds on this tradition by estimating a Roy model of occupational choice based on college major, and using it to construct a mismatch measure that adjusts for selection. We then extend this analysis into an equilibrium framework, embedding our estimates into a nested CES production function to evaluate the aggregate implications of observed mismatch, when prices adjust to changing inputs. In this sense, our paper bridges micro-level structural modeling with macro-level equilibrium evaluation, an approach that remains rare in the mismatch literature.

Finally, our work relates to the literature on the returns to college education. Although the average college premium is large, an extensive body of research shows that the return is heterogeneous (Hendricks and Schoellman, 2014; Hendricks and Leukhina, 2018). The literature also finds that returns depend on what a graduate studies (see Altonji, Arcidiacono, and Maurel, 2016 for a survey). And more importantly on where that training is used: Lemieux (2014) and Kinsler and Pavan (2015) document that workers earn more when they work in the fields related to their college major. Closest to our focus, Liu, Salvanes, and Sørensen (2016) estimate the wage premium of each major in each industry in Norway and classify a worker as mismatched if employed in one of the industries with the lowest premia for their major.

Our paper differs from this literature in three respects. First, and most importantly, we evaluate the aggregate output consequences of mismatch in general equilibrium, allowing skill prices to adjust as workers are reallocated. Second, we directly use the return to every major–occupation pair to construct the mismatch measure, rather than inferring match quality from average earnings or from comparisons of tasks and skills. Third, we correct these returns for selection into occupations using a structural Roy model. In doing so, we show how focusing solely on individual-level outcomes may overemphasize the need for policy intervention to correct mismatch in the economy. Instead, our results suggest that such policies should be evaluated against aggregate rather than individual-level estimates, and that the measured losses are concentrated among recent graduates in specific fields.

The remainder of the paper is organized as follows. Section 2 develops the skill-mismatch framework, defining the individual (micro) and aggregate (macro) measures of mismatch. Section 3 describes the ALife dataset and the estimation sample. Section 4 discusses the identification and estimation of the wage returns and the production-function parameters. Section 5

presents the baseline micro and macro results, including the decline in mismatch over the life cycle. Section 6 examines robustness and extensions. Finally, Section 7 concludes.

2 Skill Mismatch Framework

To define mismatch, we consider an economy of K intermediate firms, each identified with a unique occupation. Intermediate firms operate competitively using a CES production function and hire workers with different skills (from different college majors). Workers choose the occupation that maximizes their utility based on the Roy model. Workers differ in their acquired skills – college majors. We take these skills as given and fixed. In this setting, we define mismatch as the deviation from the efficient allocation of college majors to occupations, which maximizes the aggregate output.

2.1 Model Components

Workers Let $\mathcal{J} = (1,\ldots,j,\ldots,J)$ denote the set of college majors and $\mathcal{K} = (1,\ldots,k,\ldots,K)$ the set of occupations. After graduating from college, a worker enters the labor market endowed with major-specific human capital $\tilde{h}_i^j = h^j a_i$, where h^j is an index equal to one if the worker studied major j at college, and a_i denotes general labor efficiency units. The worker selects an occupation every period t. The model has no intertemporal decisions, thus, we omit time-subscripts for the theory.

Occupational choice follows a Roy model: worker i with major j picks an occupation after observing earnings $\{w_{i,j,k}\}$ and the non-pecuniary utility shocks $\{\eta_{i,j,k}\}$. Utility from occupation k given major j is the sum of earnings and the shock,

$$U_{i,j,k} = \ln w_{i,j,k} \; + \; \eta_{i,j,k} \quad \text{for each } j \in \{1,...,J\}. \tag{1}$$

The shocks $\eta_{i,j,k}$ capture more than individual preferences for particular occupations. They could also quantify search and informational frictions, occupational licensing barriers, and unobserved occupational amenities.[6] The more the dispersion of these shocks across occupations dominates the differences in earnings, the more the worker's choice is driven by non-pecuniary reasons, and the further the resulting allocation can depart from the one that maximizes aggregate output. Workers pick the occupation where their utility $U_{i,j}$ is maximized,

$$U_{i,j} = \max\Big\{U_{i,j,1},\ldots,U_{i,j,k},\ldots,U_{i,j,K}\Big\}. \tag{2}$$

[6]Non-pecuniary utility shocks are commonly used in the literature to capture barriers or amenities specific to occupations or regions (Birinci et al., 2025; Auerbach et al., 2025).

Wages In this economy, wages are the product of the major-specific human capital a worker supplies, $\tilde{h}_i^j$, and the price that human capital commands where it is supplied. Let $\{p_j^k\}$ collect these prices, with p_j^k the price of major-j human capital in occupation k. The price of a given major can differ across occupations, $p_j^k \neq p_j^{k'}$, because the non-pecuniary utility shocks $\{\eta_{i,j,k}\}$ could leave some occupations over-supplied with that major and others under-supplied. Earnings of an individual i with college major j working in occupation k are therefore

$$w_{i,j,k} = p_j^k \tilde{h}_i^j = p_j^k h^j a_i \tag{3}$$

Taking the logarithm of both sides of Equation (3) yields an empirical specification for the returns to college majors. Since we take this specification to panel data, we index it by year t and allow the returns to vary over time,

$$\ln w_{i,j,k,t} = \sum_{j'=1}^{J} \sum_{k'=1}^{K} \underbrace{\beta_{j',t}^{k'}}_{\ln p_{j',t}^{k'}} D_{j',k'} + \underbrace{X_{i,t}\gamma}_{\ln a_{i,t}} + \epsilon_{i,j,k,t} \tag{4}$$

where $w_{i,j,k,t}$ is annual earnings, $\beta_{j',t}^{k'}$ is the rate of return to major-j'-specific human capital in occupation k' (or the logarithm of its price, $p_{j',t}^{k'}$), $D_{j',k'}$ is a dummy variable that takes 1 if $k' = k$ and $j' = j$, and $\epsilon_{i,j,k,t}$ is the error term. General labor efficiency units are parameterized as $a_{i,t} = e^{X_{i,t}\gamma}$, where $X_{i,t}$ is a vector of individual characteristics, gender, age, and a quadratic in work experience, and γ the associated returns. In Section 4, we discuss how to estimate $\{p_{j',t}^{k'}\}$ using the Roy model of occupational choice.

Firms Production is organized in two layers. Competitive intermediate firms, one for each occupation, hire college graduates, and a competitive final good firm aggregates their output, so that the economy's production function is a nested CES over major–occupation pairs.[7] Consider the intermediate firms first. For each occupation k, we assume a CES production function that takes workers' skills, summarized by their major j, as inputs,

$$\begin{aligned} Y^k &= F^k(H_1^k, ..., H_j^k, ..., H_J^k) \\ &= \left(\alpha_1^k \left(H_1^k\right)^{\sigma_k} + ... + \alpha_j^k \left(H_j^k\right)^{\sigma_k} + ... + \alpha_J^k \left(H_J^k\right)^{\sigma_k}\right)^{\frac{1}{\sigma_k}}, \end{aligned} \tag{5}$$

[7] We focus only on the production function of college graduates. We could further assume a final goods sector aggregating university and non-university labor. This, however, will not affect the results of the analysis as long as university and non-university labor is separable and aggregated in the final CES goods sector only.

where $H_j^k = \sum_{i \in I_j^k} \tilde{h}_i$ is the supply of labor efficiency units of individuals with college major j to occupation k, with I_j^k the set of indexes for the individuals with major j in occupation k. Because each major summarizes a different bundle of skills, both the weights $\left(\alpha_1^k, \alpha_2^k, ..., \alpha_J^k\right)$ and the substitution parameter across majors, σ_k, are occupation-specific.

The competitive final good producer then aggregates occupational outputs Y^k with a CES production function,

$$\begin{aligned} Y &= F\left(Y^1, \ldots, Y^k, \ldots, Y^K\right) \\ &= \left(\alpha_1 \left(Y^1\right)^\sigma + ... + \alpha_k \left(Y^k\right)^\sigma + ... + \alpha_K \left(Y^K\right)^\sigma\right)^{\frac{1}{\sigma}}, \end{aligned} \tag{6}$$

where $(\alpha_1, \alpha_2, ..., \alpha_K)$ are the weights for all occupations and σ is the substitution parameter across occupations.[8]

Neither of the production functions in (5) or (6) carries a separate total factor productivity term. Aggregate productivity is a common scale on Y and is normalized to one. It cancels from every mismatch measure defined below, since these are all ratios. Occupation-specific productivity is absorbed into the occupational weight $\{\alpha_k\}$. In Section 4, we discuss how to estimate the parameters of the production functions, $\left\{(\alpha_1^k, \alpha_2^k, ..., \alpha_J^k)\right\}$, $\{\sigma_k\}$, $(\alpha_1, \alpha_2, ..., \alpha_K)$, and σ.

Major-specific human capital is the only input in production. Profit maximization by competitive firms therefore equates the price of major-j human capital in occupation k to its marginal product,

$$p_j^k = \frac{\partial Y}{\partial Y^k} \frac{\partial Y^k}{\partial H_j^k} = \alpha_k Y^{1-\sigma} \left(\sum_{j'=1}^{J} \alpha_{j'}^k \left(H_{j'}^k\right)^{\sigma_k} \right)^{\frac{\sigma}{\sigma_k} - 1} \alpha_j^k \left(H_j^k\right)^{\sigma_k - 1}. \tag{7}$$

2.2 Skill Mismatch Measures

We define skill mismatch both at the individual level and for the aggregate economy. First, we measure individual mismatch as wage gains by reallocating workers across occupations while keeping returns fixed. Second, we measure aggregate mismatch as output gains in a set-up where prices adjust when workers relocate across occupations.

Individual (Micro) Mismatch Let $k^*(j)$ denote the occupation in which the skills acquired in college major j earn the highest return, $p_j^{k^*(j)} = \max_k p_j^k$, which we call the *top*-occupation for

[8] Aggregating occupational output into a final good in (6) is a standard approach in the task-based literature (see, for example, Autor and Dorn, 2013).

major j.[9] We define average individual *within* major-occupation mismatch for major j as

$$m_j^{\text{micro}} = \frac{\sum_{i \in I^j} p_j^{k^*(j)} \tilde{h}_i - \sum_{i \in I^j} p_j^k \tilde{h}_i}{\sum_{i \in I^j} p_j^k \tilde{h}_i}, \tag{8}$$

where I^j is the set of individuals with major j. Holding returns fixed, m_j^{micro} is the wage gain from allocating every worker with major j to their top occupation $k^*(j)$, expressed relative to the total wages those workers currently earn.[10]

Summing across majors, economy-wide individual mismatch is

$$\mathcal{M}^{\text{micro}} = \frac{\sum_{i \in I} p_j^{k^*(j)} \tilde{h}_i - \sum_{i \in I} p_j^k \tilde{h}_i}{\sum_{i \in I} p_j^k \tilde{h}_i}, \tag{9}$$

where I is the set of all individuals in the economy. This rate represents the average potential gain if all individuals were to *switch* to their top occupation conditional on majors. By assumption, returns $\{p_j^k\}$ are fixed during the reallocation. In reality, reallocating talent across occupations changes those returns, which is what the macro measure below accounts for.

Aggregate (Macro) Mismatch We first define the efficient allocation of workers to occupations, $\hat{\mathbf{H}}^k = \left(\hat{H}_1^k, ..., \hat{H}_j^k, ... \hat{H}_J^k\right)$ for all $k \in \mathcal{K}$, as the allocation that maximizes aggregate output Y. Obtaining it jointly across occupations amounts to solving for $K \times J$ unknowns, which are pinned down by the following $(K-1) \times J + J = K \times J$ equations,

$$\frac{\hat{p}_j^{k'}}{\hat{p}_j^k} = \frac{\frac{\partial Y}{\partial Y^{k'}} \frac{\partial Y^{k'}}{\partial \hat{H}_j^{k'}}}{\frac{\partial Y}{\partial Y^k} \frac{\partial Y^k}{\partial \hat{H}_j^k}} = \frac{\alpha_{k'} \left(\sum_{j'=1}^J \alpha_{j'}^{k'} \left(\hat{H}_{j'}^{k'}\right)^{\sigma_{k'}}\right)^{\frac{\sigma}{\sigma_{k'}}-1} \alpha_j^{k'} \left(\hat{H}_j^{k'}\right)^{\sigma_{k'}-1}}{\alpha_k \left(\sum_{j'=1}^J \alpha_{j'}^k \left(\hat{H}_{j'}^k\right)^{\sigma_k}\right)^{\frac{\sigma}{\sigma_k}-1} \alpha_j^k \left(\hat{H}_j^k\right)^{\sigma_k-1}} = 1, \tag{10}$$

and

$$\hat{H}_j^1 + ... + \hat{H}_j^k + ... + \hat{H}_j^K = \bar{H}_j, \quad \forall j \in \mathcal{J}, \tag{11}$$

Equation (10) equalizes relative returns within major j across all occupations. Equation (11) is the market clearing condition, the labor allocated across occupations within each major j exhausts the total supply of that major, $\bar{H}_j$.

Equation (10) implies that in a world without non-pecuniary utility shocks, all occupations

[9] In the empirical implementation the maximum is taken over the occupations that employ at least 10 percent of the major's graduates (Section 3); Appendix E.3 reports the alternatives.

[10] Similar to our micro measure, Leighton and Speer (2023) construct a major–occupation match-quality index based on relative wages across major–occupation pairs.

pay the same conditional on major. Denote the price in the efficient allocation in the aggregate as

$$\hat{p}_j^k = \frac{\partial Y}{\partial H_j^k}\left(\hat{\mathbf{H}}^1, ..., \hat{\mathbf{H}}^K\right). \tag{12}$$

In the efficient allocation, the returns to each major are the same across occupations. Thus, we can define a unique price for each major j as

$$\hat{p}_j \equiv \hat{p}_j^1 = ... = \hat{p}_j^K. \tag{13}$$

The aggregate macro mismatch can be defined as

$$\mathcal{M}^{\text{macro}} = \frac{F\left(F^1\left(\hat{\mathbf{H}}^1\right), ..., F^K\left(\hat{\mathbf{H}}^K\right)\right) - F\left(F^1\left(\mathbf{H}^1\right), ..., F^K\left(\mathbf{H}^K\right)\right)}{F\left(F^1\left(\mathbf{H}^1\right), ..., F^K\left(\mathbf{H}^K\right)\right)}, \tag{14}$$

$$= \frac{\sum_{i \in I} \hat{p}_j \tilde{h}_i - \sum_{i \in I} p_j^k \tilde{h}_i}{\sum_{i \in I} p_j^k \tilde{h}_i}. \tag{15}$$

Similarly the macro mismatch for major j is defined as

$$m_j^{\text{macro}} = \frac{\sum_{i \in I^j} \hat{p}_j \tilde{h}_i - \sum_{i \in I^j} p_j^k \tilde{h}_i}{\sum_{i \in I^j} p_j^k \tilde{h}_i}. \tag{16}$$

These are the counterparts of Equations (8) and (9) with endogenous prices. They measure the potential gains from moving to the frictionless allocation, now evaluated at the prices $\hat{p}_j$ that prevail once workers have been reallocated.

Our macro mismatch measure thus reflects the output loss from the observed allocation of workers across occupations relative to the output-maximizing one. The two allocations differ in a single respect: in the efficient allocation the marginal product of a given major, the price of its human capital, is equalized across occupations, whereas in the observed allocation both differ across occupations. Prices move between the two because reallocating workers changes the supply of major-specific human capital within each occupation, and with it the marginal product of that major.

This measure does not, however, capture the welfare loss from mismatch. This is because the non-pecuniary utility shocks $\eta_{i,j,k}$ may reflect not only frictions in the labor markets but also occupation and/or individual-specific amenities shocks. Thus, reallocating workers to maximize output may not necessarily lead to a Pareto improvement in individual welfare. Because tastes and labor-market frictions are not separately identified within η, we interpret the

estimates as an upper bound on mismatch-induced inefficiencies.[11]

3 Data

We make use of the ALife dataset (Carter et al., 2021), a longitudinal dataset administered by the Australian Tax Office (ATO). ALife contains an extract of personal income tax records from 1991 to 2019 for a 10 percent representative sample. The data provide accurate and detailed information on variables that are required to estimate individual tax liabilities. It collates employer records, salary information, and self-reported data from taxpayers. ALife, given Australian tertiary financing policies, can also be merged with details on Higher Education loan information or the study records of terminal degrees.[12] While the tax records extend back to 1991, the linked higher education records are available only from 2003 onward, which determines the start of our estimation window. We do not extend the sample beyond 2019.[13]

Given the administrative nature of ALife, the data has very little attrition, with on average, 96.5 percent of tax filers who lodge in a given year also lodge in the subsequent income year (Carter et al., 2021). The dataset contains around 300 variables from individual tax records, including some constructed variables, such as the presence of a spouse. Demographic information includes age at 30 June of a given year, gender, residential location, occupation, and whether the tax filer is a non-resident for tax purposes. A caveat, given that occupation information, is self-reported and prefilled in later tax years when the tax filing system moved to a predominantly online reporting basis, there is potential for misreporting and limited occupation switching (Hathorne and Breunig, 2022). To ameliorate the potential noise in occupation information, we restrict our sample to college-educated individuals aged 21 to 35.[14] We also restrict the sample to the workers who only choose one specific major in college. This implicitly assumes that each individual has only one *top*-occupation which is solely determined by their unique college major.[15] We also exclude the self-employed and workers in managerial

[11] A welfare-maximizing planner would instead require separately identifying frictions and amenities within η. One approach imposes additional structure on the mapping from observed choices to frictions and amenities, for example, attributing imperfect mobility to search frictions alone and treating the residual deviation as firm- or individual-specific amenities (Taber and Vejlin, 2020; Lamadon et al., 2022; Morchio and Moser, 2024), or interpreting wedges relative to a reference group (e.g., white men) as discrimination (Hsieh et al., 2019). An alternative is to estimate job- and individual-specific amenities directly from survey data (Humlum et al., 2025) and attribute the residual deviation from the output-maximizing allocation to frictions.

[12] For additional data details see Appendix B.

[13] We stop in 2019 to avoid the COVID-19 shock to the labor market. The pandemic disrupted employment and occupational mobility in ways unrelated to major–occupation mismatch, and including those years would attribute the resulting reallocation to changes in mismatch.

[14] Hathorne and Breunig (2022) report similar findings across HILDA, a national representative sample, and ALife for the age group 25 to 38.

[15] Our results show that the proportion of workers with more than one major as classified under the 2-digit

Table 1: Summary Statistics: Estimation Sample

Panel A: Sample structure		*Panel B: Worker characteristics*			
	Number		Mean	Std. Dev.	Median
Individuals	138,113	Age	28.008	4.031	28.000
Person-year observations	641,249	Female	0.565	0.496	
Sample period	2003–2019	Real annual earnings (2019 AUD)	66,248	41,489	60,097
College majors (2-digit ASCED)	10	Log real annual earnings	10.956	0.539	11.004
Occupations (2-digit ANZSCO)	27	Work experience (years)	6.405	3.496	6.000
Regions (GCCSA)	15	Occupational tenure (years)	4.428	3.129	4.000
Major-occupation cells (≥ 100 obs.)	244	Occupation switch rate (year-to-year)	0.098	0.298	

Notes: The estimation sample covers single-major college graduates (bachelor's degree, diploma, or certificate; postgraduate qualifications excluded; graduation age 18 or older) aged 21 to 35 in tax years 2003–2019, excluding the self-employed and workers in managerial and protective-service occupations, and restricted to major-occupation cells with at least 100 observations, and to person-years with annual earnings of at least half the full-time, full-year minimum wage. Earnings are annual wage and salary income, CPI-deflated to 2019 Australian dollars. Work experience and occupational tenure are years accumulated in the panel; the occupation switch rate is the share of consecutive person-year observations with a 2-digit ANZSCO occupation different from the previous year. Regions are Greater Capital City Statistical Areas (GCCSA), the level at which the region-by-year fixed effects are absorbed. Food, Hospitality and Personal Services (ASCED 11) is excluded from the estimation sample.

and protective-service occupations, and restrict to major-occupation cells with at least 100 observations. Finally, because tax records report annual wage and salary income but not hours or weeks worked, we impose a minimum labor-market attachment: we keep person-year observations with annual earnings of at least half of what a full-time, full-year worker earns at the national minimum wage (see Appendix B).

Table 1 summarizes the estimation sample and the variables used in the wage estimation. Panel A reports the structure of the sample, 138,113 individuals and 641,249 person-year observations across 27 occupations, 15 regions, and 244 major–occupation cells. Panel B reports worker-level characteristics across log real annual earnings (the outcome), age, gender, work experience, and occupational tenure (individual controls).

We classify college majors at the 2-digit ASCED level and occupations at the 2-digit ANZSCO level. Tables A1 and A2 in Appendix A list the 10 majors and 27 occupations in the estimation sample. Table 2 summarizes occupational shares within each major: column (2) reports the number of occupations matched to each field; columns (3)–(5) the mean, minimum, and maximum occupational share within the field; and columns (6)–(8) the top occupation by major, its employment share, and its mean annual earnings. As shown there, the number of occupations varies substantially across majors. Management and Commerce and Society and Culture match to all 27 occupations, while more specialized fields match to fewer.

Observing large heterogeneity across fields of study in the number of occupational matches and employment shares, we study wage penalties from working in an occupation relative to the *top*-occupation of an individual's major. There are potentially many ways of defining the *top*-occupation. In our baseline, we use both wage and share information. We rank occupations

ASCED fields of study account for 19.6 percent of all workers.

Table 2: Occupation Share and Top Occupation by Major

		Occupational Share			Top Occupation		
2-digit ASCED Major	No.Occ	Mean	Min	Max	Top-Occupation	Share	Earnings
Natural and Physical Sciences	26	0.038	0.005	0.325	Design, Engineering, Science and Transport Professionals	0.325	76,348
Information Technology	20	0.050	0.004	0.471	ICT Professionals	0.471	87,449
Engineering and Related Technologies	24	0.042	0.002	0.582	Design, Engineering, Science and Transport Professionals	0.582	94,139
Architecture and Building	16	0.062	0.012	0.443	Engineering, Ict and Science Technicians	0.136	72,533
Agriculture, Environmental and Related Studies	23	0.043	0.008	0.349	Design, Engineering, Science and Transport Professionals	0.349	76,476
Health	27	0.037	0.001	0.654	Health Professionals	0.654	69,172
Education	27	0.037	0.002	0.696	Education Professionals	0.696	65,211
Management and Commerce	27	0.037	0.003	0.397	Business, Human Resource and Marketing Professionals	0.397	85,057
Society and Culture	27	0.037	0.005	0.240	Legal, Social and Welfare Professionals	0.240	74,620
Creative Arts	27	0.037	0.002	0.130	Business, Human Resource and Marketing Professionals	0.106	69,335

Notes: This table shows, for each field of study, the number of occupations matched to it (column 2); the mean, minimum, and maximum occupational share within the field (columns 3-5); and the field's top occupation, its share, and its mean annual earnings (columns 6-8). The product of the mean occupational share and number of occupations within the field is not 1, since the estimation sample removes field-by-occupation cells that contain less than 100 observations.

within fields of study by their selection-corrected returns subject to having at least a 10 percent share of major-specific graduates. We define the *top*-occupation as the occupation with the highest return among this set.[16] We use the 10 percent share, as the goal is not comparing mismatch relative to a *superstar*, but relative to a typical college graduate.

4 Estimation

The estimation proceeds in two steps. We first estimate the returns to college majors by occupation from a selection-corrected wage equation. We then take these returns as inputs and estimate the production function parameters from the firm's first-order conditions, exploiting variation across years and regions. Prices and quantities therefore carry time and region subscripts: $p^k_{j,r,t}$ and $H^k_{j,r,t}$ denote the price and quantity of major-j human capital in occupation k in year t and region r, and $p^k_{r,t}$ and $Y^k_{r,t}$ the occupation-level price and output. Production-function parameters are common across regions.

4.1 Returns to College Majors

Suppose that, after graduating from college, a worker enters the labor market and provides college-major-specific human capital. His/her wage equation is then

$$\ln w_{i,j,k,r,c,t} = \ln p^k_{j,t} + X_{i,t}\gamma + \lambda_{r,t} + \nu_c + \epsilon_{i,j,k,c,r,t}, \tag{17}$$

[16] We also perform robustness by: (1) ranking occupations based on the highest wage returns within majors with at least 5 percent share of graduates, and (2) based only on the highest wage returns within majors, and (3) based only on their highest share of employment within majors. Top occupations are unchanged under (1) and change for two majors under (3); under (2) they change for six majors and mismatch rises to 34 percent, driven by small, highly paid cells. See Appendix E.3 for the resulting mismatch estimates.

which extends the efficiency units of Equation (4) beyond the individual characteristics $X_{i,t}$.[17] Here $\lambda_{r,t}$ is a region-by-year productivity term, absorbing local labor market conditions that shift the earnings of all graduates in region r in year t; ν_c is a graduation-cohort productivity term, absorbing differences in the labor market a cohort enters and in the composition of graduates across cohorts; and $\epsilon_{i,j,k,c,r,t}$ is an idiosyncratic productivity shock. Absorbing regional and cohort variation in this way means the returns $p^k_{j,t}$ are identified from wage differences across major–occupation cells within a region-year and within a graduation cohort, rather than from where and when graduates happen to work.

4.1.1 Occupational Choice Model conditional on College Major

Expanding on the occupation choice model from Equation (1), individual i, who graduates in major j of cohort c, chooses occupation k in calendar year t. The overall utility of choosing occupation k given major j is determined by log earnings $\ln w_{i,j,k,c,r,t}$ and non-pecuniary utility $\eta_{i,j,k,c,r,t}$ received from unobserved occupational characteristics (i.e., working conditions, taste, etc.). The utility from Equation (1) can be rewritten in terms of a conditional mean $\tilde{U}_{i,j,k,c,r,t}$ and an individual-specific shock $e_{i,j,k,c,r,t}$ such that,

$$
\begin{aligned}
U_{i,j,k,c,r,t} &= \underbrace{\ln w_{i,j,k,c,r,t}}_{\text{Log Earnings}} + \underbrace{\eta_{i,j,k,c,r,t}}_{\text{Non-pecuniary Utility}} \qquad \text{for each } j \in \{1,...,J\} \qquad (18)\\
&= \underbrace{E[\ln w_{i,j,k,c,r,t}|W_{i,t}] + E[\eta_{i,j,k,c,r,t}|W_{i,t}]}_{\text{Conditional Mean: } \tilde{U}_{i,j,k,c,r,t}} + \overbrace{\underbrace{\epsilon_{i,j,k,c,r,t}}_{\substack{\text{Errors in}\\ \text{Earnings Equation}}} + \underbrace{\kappa_{i,j,k,c,r,t}}_{\substack{\text{Non-pecuniary Utility}\\ \text{Shock}}}}^{\text{Shock: } e_{i,j,k,c,r,t}},\\
&= \tilde{U}_{i,j,k,c,r,t} \; + \; e_{i,j,k,c,r,t}
\end{aligned}
$$

where $\tilde{U}_{i,j,k,c,r,t}$ is the utility conditional on $W_{i,t}$, which includes a vector of individual observed characteristics by calendar year t and any exclusion restrictions for identifying the occupation preference from earnings. Utility shock of occupational choice $e_{i,j,k,c,r,t}$ consists of earnings shock $\epsilon_{i,j,k,c,r,t}$ and non-pecuniary utility shock $\kappa_{i,j,k,c,r,t}$. For maximizing the utility, individual i chooses occupation k in year t if and only if $U_{i,j,k,c,r,t} \geqslant U_{i,j,k',c,r,t}$; $\forall\ k' \neq k$. We use indicator $M_{i,j,k,c,r,t}$ to denote the event that we observe an individual who chooses occupation k given

[17] University quality may also affect graduates' human capital. If university identifiers were available, we could incorporate university fixed effects into individual efficiency units and carry them across occupational reallocations. Such identifiers are unavailable in our data.

major j so that,

$$M_{i,j,k,c,r,t} = 1 \text{ iff } e_{i,j,k',c,r,t} - e_{i,j,k,c,r,t} \leqslant \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t};\ \forall\ k' \neq k. \tag{19}$$

4.1.2 Correct Selection Bias: Parametric Correction Function

Specifically, individuals are observed in an j,k cell if and only if the rule (19), conditional on major j, is satisfied. There is potential selection bias, $E[\epsilon_{i,j,k,c,r,t}|M_{i,j,k,c,r,t} = 1] \neq 0$, as the unexplained factor $\epsilon_{i,j,k,c,r,t}$ can be correlated with unobserved factors that are deterministic for the observations to appear in (j,k) cell. Directly using condition (19) to pin down the selectivity bias is infeasible because a complete specification of the joint distribution of error terms in the wage equation and error terms in K selection equations for each major j would be necessary, where K is the number of occupations for major j to choose. We follow the approach by Lee (1983), Dahl (2002) and, most recenlty, Eckardt (2026) to correct for the sample selection bias for observations in j,k cell by controlling for the correction function of selection probability, $g(\hat{p}_{i,j,k,c,r,t})$, where $\hat{p}_{i,j,k,c,r,t}$ is the estimated probability of choosing the observed occupation k conditional on graduating from major j. The function $g(x)$ has a known inverse Mills ratio of the form, $\frac{-\phi(\Phi^{-1}(x))}{x}$. Details of the derivation of the selectivity bias can be found in Appendix C.1.

The earnings model that is corrected for selection bias becomes,

$$\ln w_{i,j,k,c,r,t} = \sum_{j'=1}^{J}\sum_{k'=1}^{K} \beta^{k'}_{j',t} D_{j',k'} + X_{i,t}\gamma + \lambda_{r,t} + \nu_c + \sum_{j'=1}^{J}\sum_{k'=1}^{K} \rho^{k'}_{j',t} D_{j',k'} \cdot g(\hat{p}_{i,j,k,c,r,t}) + \xi_{i,j,k,c,r,t}$$
$$\text{where } E[\xi_{i,j,k,c,r,t}|M_{i,j,k,c,r,t} = 1] = 0, \tag{20}$$

where $w_{i,j,k,c,r,t}$ is annual earnings, $\beta^{k'}_{j',t}$ is the rate of return to major-j'-specific human capital in occupation k' (or logarithm of the price ($p^{k'}_{j',t}$) of major j' in occupation k'), $D_{j',k'}$ is a dummy variable that takes the value 1 if $k' = k$ and $j' = j$, and $\xi_{i,j,k,c,r,t}$ is the error term.[18] Finally, $g(\hat{p}_{i,j,k,c,r,t})$ is the selection correction function for individuals appearing in cell j,k. In total, we use $J \times K$ correction functions, with the effect of each one on log earnings captured by the j,k specific parameter, $\rho^{k}_{j,t}$.[19]

As the occupation choice set varies considerably across majors (see Table 2), to make estimation feasible, we reduce the number of correction functions into $J \times 10$. Specifically, for all

[18] The returns used for the micro estimates are estimated in a single regression pooled over 2003–2019 (Section 5.1); for the production-function estimation, we re-estimate them separately for each region and year bin (Appendix D).

[19] Appendix E.2 instead treats the unobserved ability behind selection as part of the worker's human capital rather than as a bias.

individuals with major j, we partition the entire observed (major-specific) occupational set into ordered 10 sets based on the descending order of occupational share within the major: with (1) the first set comprising the *top*-occupation for major j (i.e., the occupation with the highest wage conditional on a 10 percent share of the major j); and (2) the remaining nine even sets of occupations for major j, ordered with a decrease of the occupational share to major j.[20] We estimate the individual's occupational choice probabilities conditional on the field of study j with the individual characteristics and exclusion restrictions, local labor-demand shifters for each occupation set, by a random forest algorithm (see Appendix C.2 for more detail).

4.2 Estimating Production Function Parameters

To solve for the frictionless allocation through Equations (10) and (11), and to define the mismatch measures through Equation (14), we need to know the parameter values of the production function. We first derive the reduced-form expressions from the first-order conditions of the representative firm profit maximization problem. We then exploit the region-time variations to estimate the production function parameters. We estimate two specifications. In the baseline, all production-function parameters are constant over time, and only prices and quantities vary across regions and year bins. In the technical-change specification used in Section 5.3, the major weights $\{\alpha^k_{j,t}\}$ and the occupation weights $\{\alpha_{k,t}\}$ follow linear trends in log ratios, while the substitution parameters σ_k and σ are held constant.

4.2.1 Deriving the Reduced Form Expressions

We take the first-order condition for Equation (5) that yields,

$$\frac{p^k_{j',r,t}}{p^k_{j,r,t}} = \frac{\partial F^k / \partial H^k_{j',r,t}}{\partial F^k / \partial H^k_{j,r,t}} = \frac{\alpha^k_{j',t}}{\alpha^k_{j,t}} \left(\frac{H^k_{j',r,t}}{H^k_{j,r,t}} \right)^{\sigma_k - 1}. \tag{21}$$

Taking the logarithm on both sides yields,

$$\underbrace{\ln\left(p^k_{j',r,t}/p^k_{j,r,t}\right)}_{\beta^k_{j',r,t} - \beta^k_{j,r,t}} = \ln\left(\alpha^k_{j',t}/\alpha^k_{j,t}\right) + (\sigma_k - 1)\ln\left(H^k_{j',r,t}/H^k_{j,r,t}\right). \tag{22}$$

[20] Occupations are approximately evenly placed in each set based on the descending order of occupational share within the major, so that the second occupation set is more major-related than the third set, and so forth. We also perform robustness exercises with 8 and 12 occupation groups, respectively. Results are similar across the three versions. The ten sets only restrict the coefficient ρ to be common within each major-by-set group; the choice probability entering the correction remains the predicted probability of the worker's observed occupation.

Equation (22) is the first equation we estimate with the data.

Next, the first-order condition for Equation (6) is,

$$\frac{p_{r,t}^{k'}}{p_{r,t}^{k}} = \frac{\partial F/\partial Y_{r,t}^{k'}}{\partial F/\partial Y_{r,t}^{k}} = \frac{\alpha_{k',t}}{\alpha_{k,t}} \left(\frac{Y_{r,t}^{k'}}{Y_{r,t}^{k}} \right)^{\sigma-1}. \tag{23}$$

By taking the logarithm of Equation (23), we obtain,

$$\ln\left(\frac{p_{r,t}^{k'}}{p_{r,t}^{k}} \right) = \ln\left(\frac{\alpha_{k',t}}{\alpha_{k,t}} \right) + (\sigma - 1) \ln\left(\frac{Y_{r,t}^{k'}}{Y_{r,t}^{k}} \right), \tag{24}$$

where $Y_{r,t}^{k}$ is regional output given by Equation (5). Using the equilibrium profit condition, $p_{r,t}^{k} Y_{r,t}^{k} = \sum_j p_{j,r,t}^{k} H_{j,r,t}^{k}$, we solve for the price of occupation k,

$$p_{r,t}^{k} = \left(\sum_j \left(\alpha_{j,t}^{k} \right)^{-\frac{1}{\sigma_k - 1}} \left(p_{j,r,t}^{k} \right)^{\frac{\sigma_k}{\sigma_k - 1}} \right)^{(1 - \frac{1}{\sigma_k})}. \tag{25}$$

Substituting Equation (25) for Equation (24) yields,

$$\ln\left(\frac{\left(\sum_j \left(\alpha_{j,t}^{k'} \right)^{-\frac{1}{\sigma_{k'} - 1}} \left(p_{j,r,t}^{k'} \right)^{\frac{\sigma_{k'}}{\sigma_{k'} - 1}} \right)^{(1 - \frac{1}{\sigma_{k'}})}}{\left(\sum_j \left(\alpha_{j,t}^{k} \right)^{-\frac{1}{\sigma_k - 1}} \left(p_{j,r,t}^{k} \right)^{\frac{\sigma_k}{\sigma_k - 1}} \right)^{(1 - \frac{1}{\sigma_k})}} \right) = \ln\left(\frac{\alpha_{k',t}}{\alpha_{k,t}} \right) + (\sigma - 1) \ln\left(\frac{Y_{r,t}^{k'}}{Y_{r,t}^{k}} \right). \tag{26}$$

This is the second equation we estimate.

As discussed in Section 2, the weight $\{\alpha_{k,t}\}$ already absorbs occupation-specific productivity, so the two cannot be told apart in (26); this is not crucial for our exercise, since only the composite enters the marginal products (7) that determine the frictionless allocation. A typical assumption in the literature (see Katz and Murphy (1992)) is to assume a linear time trend in technological change, e.g., the expression can be approximated by,

$$\ln\left(\frac{\alpha_{k',t}}{\alpha_{k,t}} \right) = \gamma_{0,kk'} + \gamma_{1,kk'} t,$$

when using time-series data or simply $\gamma_{0,kk'}$ if using only cross-sectional variation. Note, throughout the analysis we assume that the production technologies are the same across all regions in Australia, but potentially vary over time.

4.2.2 Exploiting Time and Regional Variations

For the occupation-specific production functions, we estimate parameters $\{\alpha^k_{j,t}\}$ and $\{\sigma_k\}$ using variation in $p^k_{j,r,t}$ and $H^k_{j,r,t}$ across regions and year bins, where the region-specific returns $p^k_{j,r,t}$ come from re-estimating the earnings equation separately in each region and year bin (Appendix D). For the aggregate production function, we estimate $\{\alpha_{k,t}\}$ and σ using variation in $Y^k_{r,t}$ and $p^k_{r,t}$. For the baseline estimation, the production-function parameters are fixed across regions and time, while prices and quantities vary across regions and year bins.

Due to the sampling limitations, we group years into 7 year groups to estimate Equations (22) and (26). The seven groups are 2003–2005, 2006–2007, 2008–2009, 2010–2012, 2013–2015, 2016–2017, and 2018–2019. Therefore, both micro and macro evolution in mismatch are computed by Equations (9) and (14) over 7 year groups for comparison purposes. Throughout, the time subscript t indexes these year groups (year bins) rather than single calendar years. We leave further detail of parameters estimation to Appendix D. Table D1 in Appendix D shows the estimation results for occupation-specific production function parameters. Our results on the elasticity of substitution between occupations, $\frac{1}{1-\sigma} = 2.332$, are comparable to Burstein et al. (2019) and Caunedo et al. (2023), who estimate a model with capital.[21]

5 Results

This section presents the paper's two main findings. First, the cost of mismatch depends critically on the equilibrium concept: under fixed returns the average worker loses 27.1 percent of wages to mismatch, but once prices respond to the planner's reallocation, the aggregate output loss is 9.8 percent, roughly one-third of the micro estimate. Both figures are gains relative to the observed allocation, as defined in Equations (9) and (14); expressed as shortfalls relative to the efficient benchmark they are 21.3 and 8.9 percent. Ignoring equilibrium price adjustment therefore leads to a substantial overstatement of the aggregate cost of mismatch, with direct implications for policies calibrated to individual-level estimates. Second, mismatch is concentrated among young workers. Losses decline steeply over the life cycle and, in aggregate, largely disappears by age 30. Section 5.1 establishes the first result, Section 5.2 the second, and Section 5.3 tracks how mismatch has evolved since 2003.

[21] Burstein et al. (2019) report estimates of $\frac{1}{1-\sigma}$ ranging from 1.78 to 3.27 across specifications; our estimate lies in this range. We conduct a further robustness check using the estimated elasticity of Caunedo et al. (2023), $\frac{1}{1-\sigma} = 1.34$.

5.1 Base Estimates

We begin with the micro estimates.[22] Table 3 reports average major-specific mismatch based on Equation (8) and overall mismatch in the aggregate economy based on Equation (9) for the average worker from 2003 to 2019, with standard errors calculated using the delta method.[23] Columns (3) and (4) show mismatch based on OLS and selection-corrected returns, respectively – the estimates of β_j^k from Equation (20).[24] Correcting for selection raises average mismatch most for the STEM-related fields (Natural and Physical Sciences, Information Technology, and Engineering and Related Technologies), Management and Commerce, Society and Culture, and Creative Arts, and it changes the ordering across majors: Engineering rises from below the economy-wide OLS rate to among the highest, while Architecture and Building, Agriculture, and Health are little changed or fall. Aggregating across majors, the baseline mismatch under fixed returns, calculated using Equation (9), is 27.1 percent. In summary, holding returns fixed, the average worker could raise their wage by more than a quarter in moving to the occupation where their major commands the highest return.

Column (5) of Table 3 reports the corresponding aggregate mismatch by major, with bootstrap standard errors in parentheses (see Appendix D.1). The contrast with column (4) constitutes the paper's main quantitative finding. Aggregating across majors, the output loss is 9.8 percent, about one-third of the 27.1 percent individual figure. This difference reflects price adjustment. As the planner reallocates workers, returns fall in the occupations that receive them and rise in those that lose them, so much of the individual wage gain from reallocation dissipates in the aggregate. The ranking of majors is largely preserved. The fields that exhibit the largest mismatch under fixed returns (Management and Commerce, Society and Culture, Creative Arts, and several STEM fields) still contribute the bulk of the aggregate loss, but each group's contribution shrinks substantially once occupation-level returns respond. The next two subsections examine how this gap between individual and aggregate mismatch varies over the life cycle and over time.

The aggregate figure is smaller than the individual one for two possible reasons. The macro measure values a different reallocation, the planner's efficient allocation rather than a move of every worker to their top occupation, and it values it at different prices, the equilibrium prices

[22] Although the theory allows major–occupation returns to vary over time, the reported micro estimates use returns estimated from a single regression that pools observations from 2003 to 2019.

[23] Specifically, mismatch estimates in Equations (8) and (9) are functions of vector $\boldsymbol{\beta}$, i.e., $h(\boldsymbol{\beta})$, where $\boldsymbol{\beta} = (\beta_j^1, ..., \beta_j^{k^*}, ..., \beta_j^k)$. Based on the multivariate delta method, we compute the variance of $h(\boldsymbol{\beta})$ using, $Var(h(\boldsymbol{\beta})) = \nabla h(\boldsymbol{\beta})^T Cov(\boldsymbol{\beta}) \nabla h(\boldsymbol{\beta})$. To reduce the computational burden, we consider $\tilde{h}_i$ to be constant.

[24] In Appendix A we report all wage differences, βs, relative to the top-match occupation across college majors (see Figure A1). Based on these major-occupation returns, Section D, details the estimation procedure for the production parameters and Table D1 provides the estimated parameters.

Table 3: Micro and Macro Mismatch by Major

		Individual (Micro)		Aggregate (Macro)
ASCED 2-digit	Major	OLS	Correction	
(1)	(2)	(3)	(4)	(5)
1	Natural and Physical Sciences	0.150 (0.004)	0.219 (0.019)	0.069 (0.007)
2	Information Technology	0.133 (0.003)	0.299 (0.016)	0.091 (0.005)
3	Engineering and Related Technologies	0.107 (0.001)	0.302 (0.010)	0.087 (0.003)
4	Architecture and Building	0.071 (0.010)	0.052 (0.020)	0.057 (0.013)
5	Agriculture, Environmental and Related Studies	0.152 (0.005)	0.147 (0.021)	0.120 (0.016)
6	Health	0.056 (0.001)	0.054 (0.004)	0.053 (0.003)
7	Education	0.076 (0.001)	0.118 (0.005)	0.065 (0.004)
8	Management and Commerce	0.151 (0.001)	0.441 (0.010)	0.141 (0.004)
9	Society and Culture	0.199 (0.003)	0.390 (0.019)	0.097 (0.005)
10	Creative Arts	0.237 (0.007)	0.380 (0.018)	0.224 (0.021)
	Economy-wide	0.127 (0.001)	0.271 (0.004)	0.098 (0.003)

Notes: This table shows the individual mismatch for each major. Columns (3) and (4) show the micro (fixed-price) mismatch estimates from the OLS and the selection-corrected estimation approaches, respectively, with standard errors in parentheses computed using the delta method. Column (5) shows the macro (equilibrium-price) mismatch, with standard errors in parentheses from the bootstrap (same draws as Figure 1). The last row shows the economy-wide mismatch.

after the reallocation rather than the observed ones. To separate the two, we value the planner's reallocation at the prices that prevail before it, $\sum_{j,k} p_j^k(\hat{H}_j^k - H_j^k)/\sum_{j,k} p_j^k H_j^k$. Because returns equal marginal products at the observed allocation and output is concave in the allocation (Appendix D), this fixed-price value is an upper bound on the output gain from the same reallocation. Table 4 reports the result on the footing of Table 3: the planner's reallocation is worth 24.5 percent at pre-reallocation prices, close to the 27.1 percent micro figure, and 9.8 percent once prices adjust. The choice of benchmark allocation therefore contributes almost nothing to the gap between the individual and aggregate measures; the equilibrium response of prices accounts for nearly all of it. Section 6.2 examines how sensitive the aggregate loss is to the substitutability of majors within and across occupations.

Table 4: Decomposing the Gap between Micro and Macro Mismatch

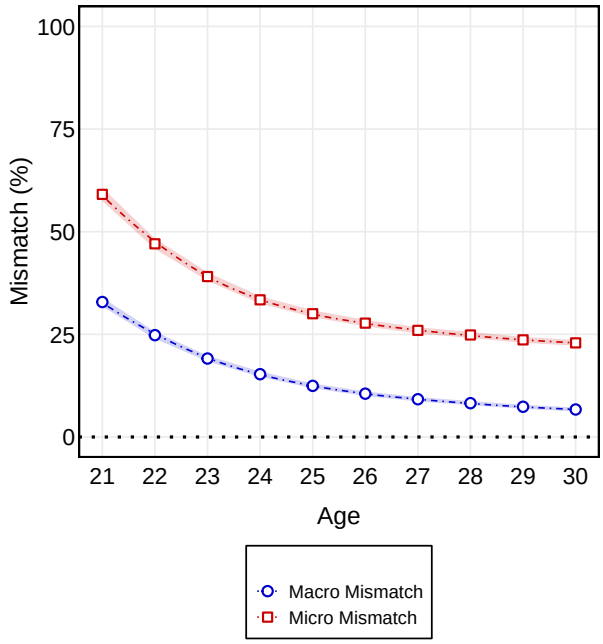
Measure	Reallocation	Prices	Gain (%)
Micro mismatch, Equation (9)	Each worker to the top occupation	Observed, p_j^k	27.1
Planner's reallocation at fixed prices	Planner's allocation $\hat{\mathbf{H}}$	Observed, p_j^k	24.5
Macro mismatch, Equation (14)	Planner's allocation $\hat{\mathbf{H}}$	Equilibrium, $\hat{p}_j$	9.8

Notes: Economy-wide gains relative to the observed wage bill, pooled over 2003–2019. Rows 1 and 3 are the economy-wide entries of Table 3, columns (4) and (5). Row 2 values the planner's reallocation at the cell prices of the observed allocation, averaged over year bins; by concavity it is an upper bound on row 3.

5.2 Life-Cycle Skill Mismatch

A natural question is whether the 27.1 percent wage loss represents a permanent burden or is concentrated at the start of working life. The distinction matters for the design of talent-reallocation and retraining policies. That is, the appropriate interventions differ depending on whether mismatch persists throughout careers or workers gradually sort into better matches. We therefore trace both mismatch measures over the life cycle.

Figure 1: Life-cycle Mismatch: Micro and Macro Estimates



Notes: This figure plots individual and aggregate mismatch using the Equations (9) and (16) over the life cycle. Shaded bands are 95% confidence intervals. The individual band is computed by the delta method, and the aggregate band by bootstrap (Appendix D.1).

We compute mismatch within each college major by age in two ways: (i) under fixed prices (micro), using Equation (8), and (ii) under flexible prices (macro), using Equation (16) with the equilibrium prices from Equation (13) after the reallocation of workers.[25] The micro age profile uses the pooled baseline returns, held fixed across ages. Recall that over the same period the average individual mismatch is 27.1 percent, while the corresponding aggregate mismatch, as defined in Equation (14), is 9.8 percent.[26] We report the 95% confidence intervals using an analytic delta-method band for the individual life-cycle series (the shaded band in

[25] Specifically, for the micro measure we restrict $i \in I_t^j$ to workers $i \in I_{a,t}^j$ who studied major j and worked at age a; we analogously restrict the macro measure to workers observed at age a. The macro profile values the efficiency units of the workers observed at age a at the equilibrium prices $\hat{p}_j$ and p_j^k of the full-sample reallocation, which are held fixed across ages. It therefore shows how the aggregate gain from reallocation is distributed across ages.

[26] These baseline estimates hold the production-function parameters fixed over time; Section 5.3 uses a specification in which the weights follow linear trends. For details on the estimation procedure, see Appendix D.

Figure 1) and a parametric bootstrap of the estimated wage returns for the production-function parameters in Table D1. Appendix D.1 describes both procedures.

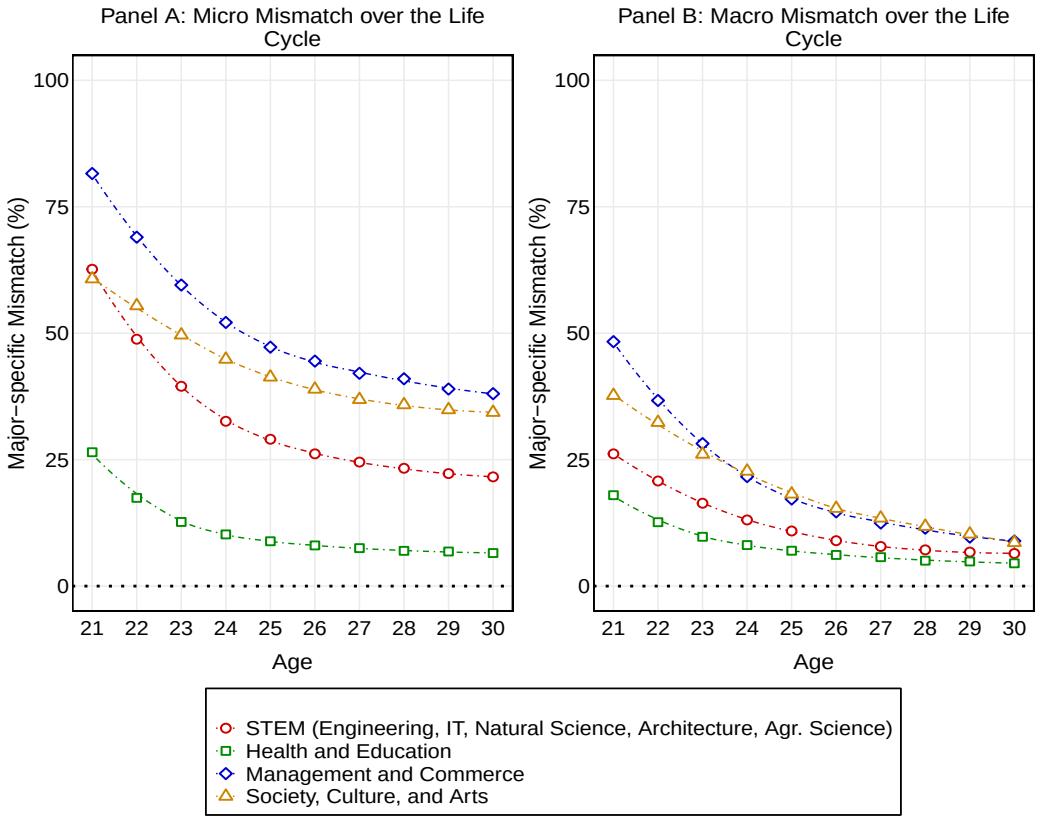
Figure 1 displays aggregate life-cycle mismatch, and Figure 2 disaggregates it into four major groups, with Panel A reporting mismatch under fixed returns and Panel B with price adjusted results.[27] The most salient pattern is a steep decline with age. Under fixed returns, mismatch roughly halves by age 30 but remains above 20 percent economy-wide; with adjusting prices it falls below 10 percent in every major group, and the aggregate output loss largely vanishes by age 30. Since mismatch is flat between age 30 and 35, we cap all figures at age 30. Under fixed prices (Panel A), STEM; Management and Commerce; and Society, Culture, and Arts majors start with the highest mismatch at age 21 (typical graduation age) and experience the steepest declines. Health and Education majors constitute an exception in both respects. Their mismatch is consistently less than half that of Commerce, Social Science, and Arts majors, and it declines only gradually over the life cycle. The second pattern is the aggregate gap documented above, now visible at every age. Comparing Panels A and B, micro mismatch is roughly 25 percentage points higher at most ages for STEM; Management and Commerce; and Society, Culture, and Arts majors under fixed returns. For Health and Education the two series are much closer.

Part of the decline is attributable to improved sorting in the labor market. All panels of Figure 3, which plot the share of individuals working in their top-occupation, show this share rising with age in every major group, consistent with previous evidence on US data (Guvenen et al., 2020). Deming and Noray (2020) similarly document that the labor market position of college graduates evolves rapidly early in the career, with the high initial returns to applied majors eroding as job skill requirements change. The shares are lowest in Commerce, Social Science and Arts and highest in Health and Education, consistent with a concept of general versus specific skills. Appendix Figure A4 shows where the remaining gap lies and what drives difference across fields of study. Comparing the observed occupational distribution of young and old workers in each field of study against the efficient aggregate allocation a number of facts emerge. First, older workers appear to more strongly represented in the top-occupation. Second, fields with low individual and aggregate mismatch exhibit less reallocation from the data to the optimum. Lastly, both in the data and in the optimum the field with small individual mismatch have a considerable larger employment share in the top-occupation.

Two implications follow. First, estimates of skill mismatch that ignore price effects due

[27] For readability of the figure, we aggregate the 10 ASCED 2-digit majors in the estimation sample into four major groups: (1) STEM-related majors include Natural and Physical Sciences, Information Technology, Engineering and Related Technologies, Architecture and Building, and Agriculture, Environmental and Related Studies; (2) Health and Education; (3) Management and Commerce (as original); and (4) Society, Culture, and Arts includes Society and Culture, and Creative Arts.

Figure 2: Individual and Aggregate Major-Specific Mismatch over the Life-Cycle



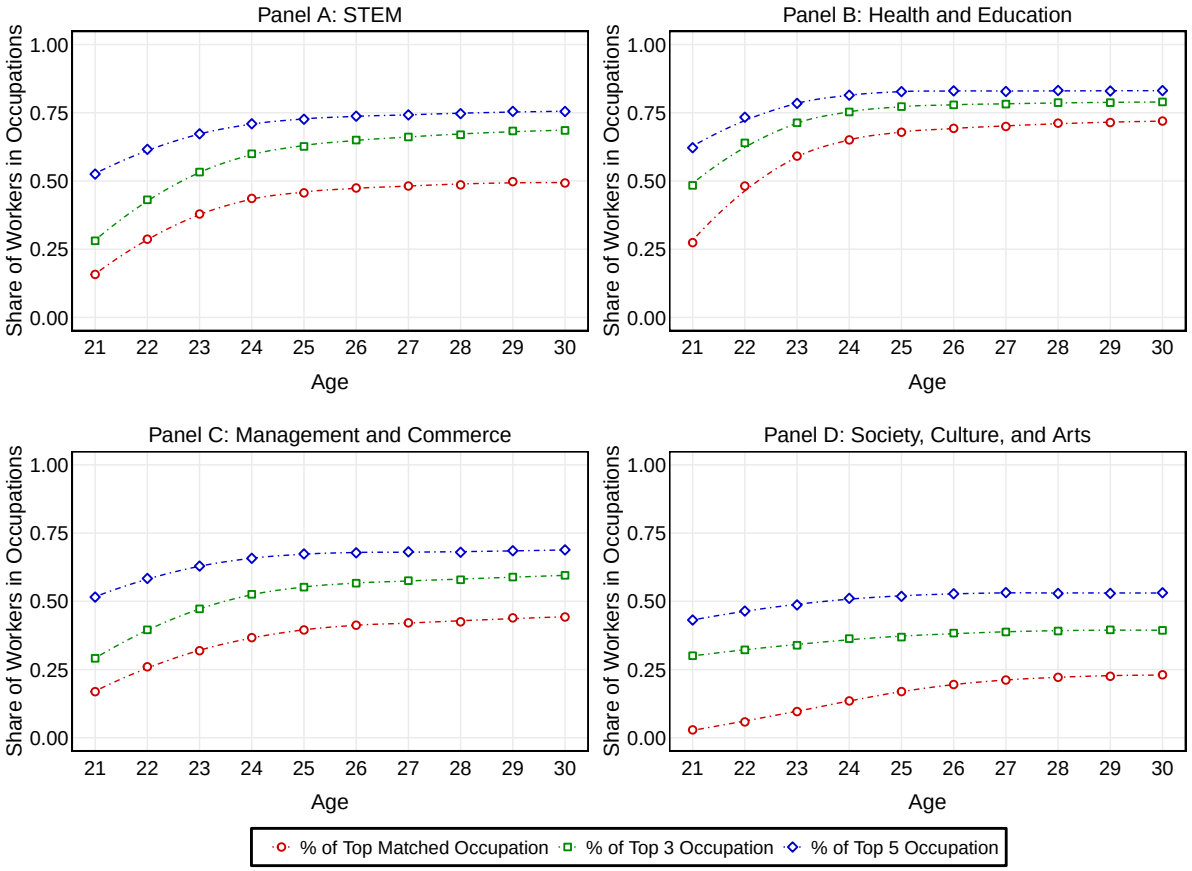
Notes: Panel A and Panel B plot micro and macro mismatch for aggregate fields of study using the Equation (9) over the life-cycle.

to reallocation of inputs will be overstated. Second, mismatch is primarily a young-worker phenomenon. The cost of mismatch essentially disappears with age once equilibrium price adjustments are taken into account. Policies concerned about skill mismatch should specifically target young workers as for some groups mismatch persists even in our macro estimates.

5.3 Evolution of Aggregate Mismatch

The results so far are averages over 2003–2019. This section briefly examines the time series. Panel A of Figure 4 plots the evolution of aggregate mismatch for the average worker under fixed returns (Equation (9)) and with adjusting prices (Equation (14)) at each year group. Averaged over the seven year bins, the output loss is 12.0 percent, above the pooled estimate of 9.8 percent in Table 3, which applies prices averaged over the year bins to the pooled allocation. Both series are relatively stable until 2012 and rise thereafter. The post-2012 increase is similar in size for the two series, about nine percentage points in each case, so price adjustment dampens the level of mismatch far more than its recent rise; from its 2010–2012 value the output

Figure 3: Share of the Top Occupation within Major over the Life-Cycle



Notes: This figure plots the share of top-matched occupation, top 3 and top 5 occupations within each field of study over the life cycle.

loss roughly doubles.

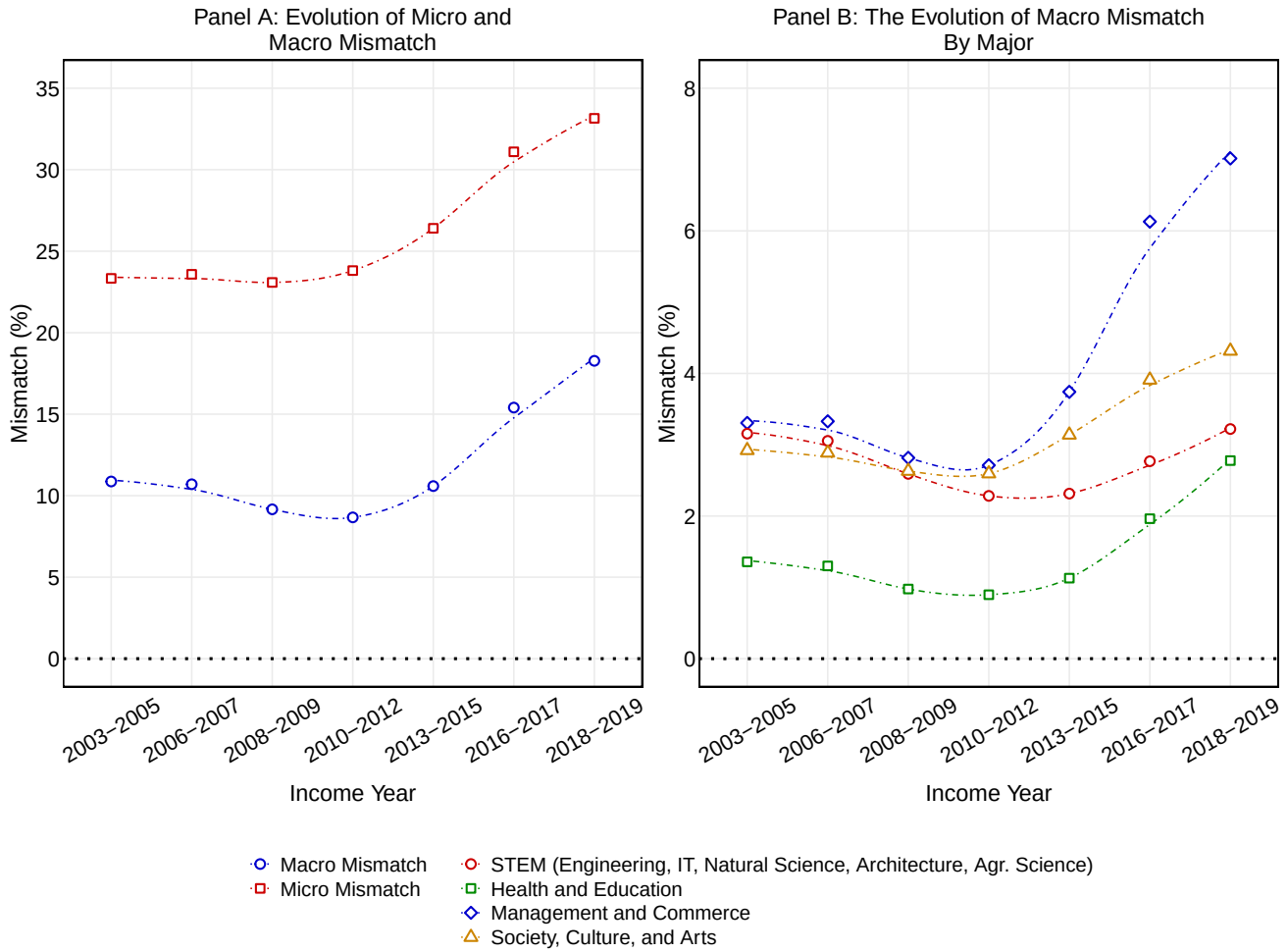
Panel B of Figure 4 decomposes the output loss by major group.[28] All major groups follow the aggregate pattern, a flat or slightly decreasing trend in output loss until 2012, rising (in some cases sharply) thereafter, with Management and Commerce exhibiting both the largest mismatch pre-2012 and the largest increase post-2012. Overall, aggregate mismatch with price adjustments is mainly driven by Management and Commerce majors (4.1 percent),

[28] We decompose the aggregate mismatch for major j in terms of GDP per capita loss by:

$$m_{j,t}^{\text{macro}} = \frac{F\left(F^1\left(\hat{\mathbf{H}}_t^1\right),...,F^K\left(\hat{\mathbf{H}}_t^K\right)\right) - F\left(F^1\left(\hat{\mathbf{H}}_{-j,t}^1\right),...,F^K\left(\hat{\mathbf{H}}_{-j,t}^K\right)\right)}{F\left(F^1\left(\hat{\mathbf{H}}_{-j,t}^1\right),...,F^K\left(\hat{\mathbf{H}}_{-j,t}^K\right)\right)} \tag{27}$$

where $\hat{\mathbf{H}}_{-j,t}^k = \left(\hat{H}_{1,t}^k,...,H_{j,t}^k,...\hat{H}_{J,t}^k\right)$ means that all majors are at their efficient allocation across occupations, except for major j remaining at their current allocation. The numerator is thus the additional output obtained by also reallocating major j, once every other major has already been reallocated, and the denominator is the output produced when major j alone remains at its observed allocation.

Figure 4: The Evolution of Mismatch over Time



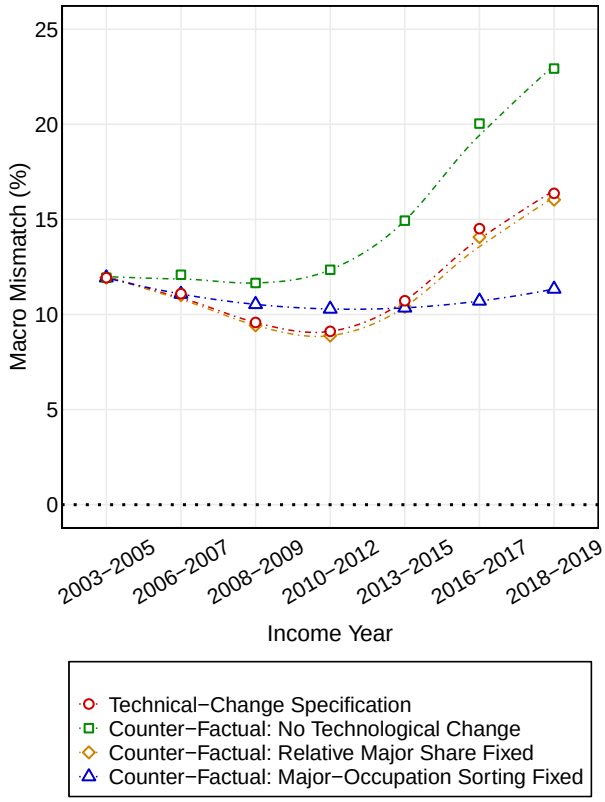
Notes: Panel A plots the evolution of estimated output loss with fixed and variable prices from 2003-2005 to 2018-2019 computed by Equations (9) and (14), respectively. Panel B plots the evolution of estimated output loss due to misallocation disaggregated into four major groups from 2003-2005 to 2018-2019. Each major group mismatch is computed by Equation (27) for each year group.

Society, Culture, and Arts majors (3.2 percent), and STEM majors (2.8 percent), while Health and Education contributes roughly a third to a half that of the other groups (1.5 percent).

We identify two possible forces which could drive the rise post-2012: (i) a change in the relative supply of college graduates across majors (see Figure A2 for the number of new college graduate workers by major over time), or (ii) a change in the efficiency of matching of workers to top occupations (see Figure A3 for the life cycle of mismatch across cohorts). The timing of the kink coincides with a major policy change. In 2009 the federal government decided to uncap Commonwealth Supported Places (CSPs), university places partly funded by the government, whose number by field of study it had previously set each year. This policy was implemented gradually from 2010 until 2012, by when every admitted domestic student

could access a subsidized university spot. Previous studies suggest the policy raised CSPs by 31.5 percent from 2009 to 2016 (Department of Education and Training, 2017). As seen in Figure A2 the resulting increase in admissions was uneven across majors and universities (not pictured) and occurred right after the Financial Crisis.

Figure 5: Counterfactual Analysis

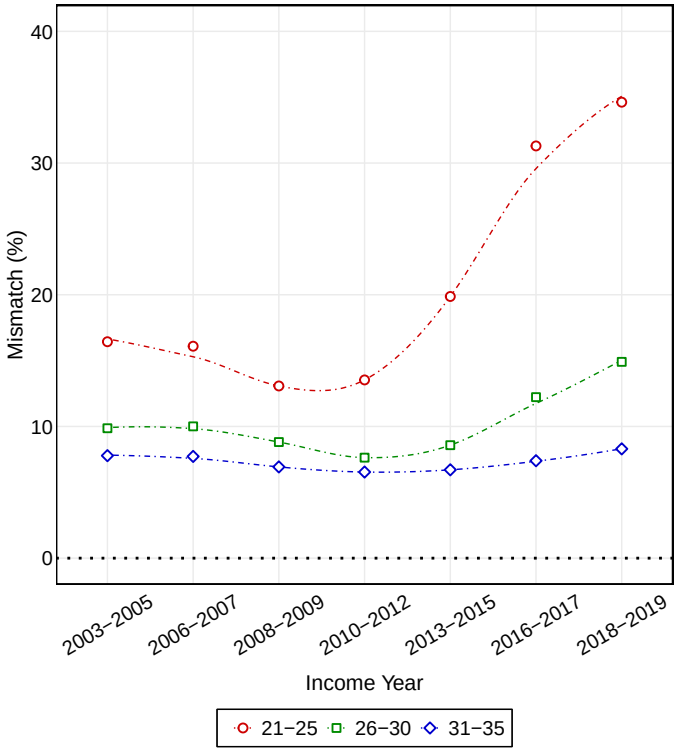


Notes: The figure shows three counterfactuals, to study the rise in the evolution of aggregate skill mismatch. The red circular line plots mismatch under the technical-change specification, in which the α's follow linear trends (Appendix D); the baseline series in Figure 4 holds them fixed. The green square lines shows the counterfactual when fixing α's to the base year, 2003-2005. The yellow diamond line fixes relative major supplies to 2003-2005 and, finally, the blue triangle line fixes the sorting of workers to occupations-by-major to the 2003-2005 base year.

We quantify the role of (i) relative supplies and (ii) sorting through two counterfactuals within our aggregate framework, comparing each year-bin post-2012 with the base period, the first year bin (2003–2005). The first counterfactual, exploiting the fact that the production function is homogeneous of degree one, keeps relative major shares at their base-period values while workers sort according to post-2012 labor market conditions: the optimal reallocation of counterfactual talent $\hat{H}^{c}{}^{k'}_{j,t}$ at time t satisfies Equations (10) and (11), and counterfactual mismatch is computed through Equation (14). The resulting series (yellow-diamond line in Figure 5) shows how output loss from mismatch falls once relative supplies across majors are kept constant. The second counterfactual instead keeps the relative occupation shares conditional

on major at their base-period values while using graduation shares from the current period; the resulting series (blue-triangle line) shows how output loss from mismatch falls once new cohorts sort into top-occupations at the same rate and timing as cohorts in the base period. Comparing the two, virtually all of the change since 2003 can be attributed to sorting.

Figure 6: Decomposition by Age Group



Notes: The figure plots the decomposition of aggregate mismatch by age groups of 21-25, 26-30, and 31-35 over years. Specifically, we compute the mismatch measure by using Equation (14) for each age group and year-bin.

Two further exercises consider whether this sorting channel reflects technology and whether it affects all workers equally. First, fixing the α's at their 2003-2005 values, shutting down any time trend in technology, leads to an even larger increase in skill mismatch (green-square line in Figure 5). The trend in the weights therefore does not account for the rise. Because the counterfactual re-evaluates the observed allocations under fixed technology rather than generating occupational choices in the absence of technical change, it does not rule out that technical change shaped the sorting changes themselves. Second, decomposing the evolution of skill mismatch by age group (21-25, 26-30, and 31-35; Figure 6) shows that mismatch is flat over the entire period for workers aged 31-35. Almost all of the increase is driven by the youngest cohort (workers aged 21-25), consistent with the life-cycle results of Section 5.2.

6 Robustness and Extensions

This section examines the robustness of the baseline estimates and develops three extensions. Section 6.1 incorporates occupational tenure and learning-by-doing, showing that both main results are robust once occupation-specific human capital is taken into account. Section 6.2 examines how sensitive the aggregate output loss is to the within- and between-occupation elasticities of substitution. Section 6.3 asks which skills cause mismatch, decomposing the wage losses into O*NET knowledge classes.

6.1 Occupational Tenure and Learning by Doing

A natural concern with the life-cycle results is that part of what we measure as mismatch may be eliminated through on-the-job learning. Graduates who start in an occupation that does not match their acquired skills may acquire the necessary skills with experience. Early mismatch thus reflects human capital still being accumulated rather than a permanent misallocation. To address this, we let major–occupation returns depend on occupational tenure $\tau_{i,j,k,t}$ in a quadratic form, extending Equation (4) to

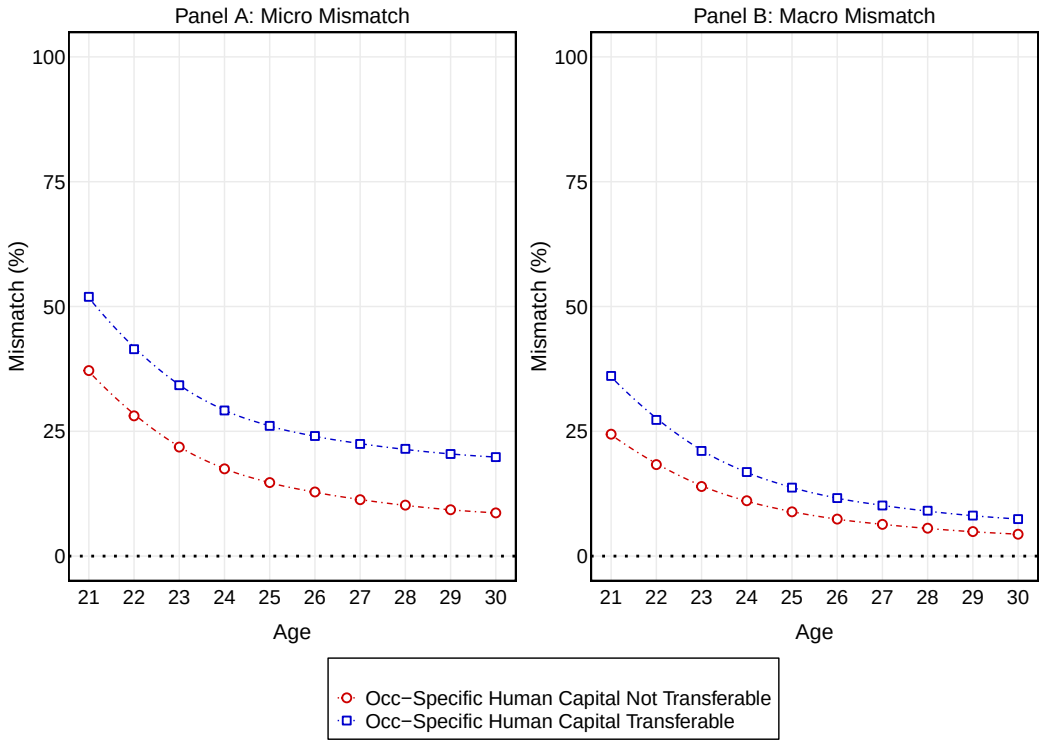
$$\ln w_{i,j,k,t} = \sum_{j'}^{J} \sum_{k'}^{K} \underbrace{\left(\theta_{j',k',t} + \eta_{1,j',k'}\tau_{i,j',k',t} + \eta_{2,j',k'}\tau_{i,j',k',t}^2\right)}_{\beta_{j',t}^{k'}} D_{j',k'} + \ln(a_{i,t}) + \epsilon_{i,j,k,t}. \tag{28}$$

Two alternative assumptions about the attribution of estimated tenure returns provide bounds on life-cycle mismatch. The *lower bound* treats returns to occupation-specific tenure as non-transferable, occupation-specific human capital (a worker who switches loses them), while the *upper bound* credits them as transferable, general human capital. The aggregate efficient allocation, in which the planner internalizes that reallocated workers arrive at their destination occupation without tenure, is derived in Appendix E.1. Intuitively, when occupation-specific human capital is valuable, the gains from reallocation shrink, so measured mismatch falls both for the individual and aggregate estimates.

Figure 7 shows that both main results are robust to this extension. Panel A plots the two fixed-price bounds. All decline steeply over the life cycle, the lower bound from about 37% at age 21 to roughly 8% by age 35, and the upper bound from about 52% to 19%. Panel B shows the output loss counterpart under the planner specification of Appendix E.1. Under both tenure valuations, mismatch declines from about 24–36% at age 21 to about 4–7% by age 30, compared to the baseline which declines from 33% to 7%. Under the forfeiture valuation, learning-by-doing accounts for part of what the baseline attributes to mismatch, but the sharp

decline with age remains under both. The contrast between the micro and macro estimates also remains. At every age and under either tenure valuation, macro mismatch is well below its micro equivalent, roughly two-thirds at age 21 (24–36% versus 37–52%) and half or less by age 30 (4.4–7.4% versus 8.7–20%). Thus, price adjustment continues to absorb most of the individual-level wage losses once occupation-specific human capital is taken into account.

Figure 7: Life-cycle Mismatch Accounting for Occupational Tenure Effects



Notes: Panel A plots the micro mismatch by age under two bounds on occupational tenure: (i) the *lower bound* treats returns to tenure as mismatch (workers forfeit tenure upon switching); (ii) the *upper bound* treats them as genuine human capital. Panel B reports the corresponding macro life-cycle series under the same two tenure valuations (Appendix E.1).

6.2 Varying the Elasticity of Substitution

The aggregate loss depends on how substitutable majors are within occupations. When majors are complements, an occupation's output is held back by the fields it lacks, so the observed allocation produces less relative to the efficient one and the gain from reallocation is larger; when they are close substitutes, the major composition of an occupation matters less for output, and so does misallocation. The productivity weights, $\{\alpha^k_{j,t}\}$, in contrast, guarantee that even with high estimated elasticities some major-occupation combinations are relatively (un)productive. For example, for *Health Professionals*, a *Information Technology* major is not observed in the estimation sample and a *Natural and Physical Science* major has a productivity weight close to zero

(0.04), but a *Health* major has a large productivity weight (0.75). The planner therefore cannot allocate *Information Technology* majors to this occupation, and the low weight makes it costly to allocate *Natural and Physical Science* majors there, although the allocation shares themselves also depend on the elasticities, the occupation weights, and each major's total supply. This section examines how sensitive the aggregate loss is to the two elasticities.

Table 5 reports the aggregate output loss with variable prices for a 3×3 grid of elasticities. The between-occupation parameter σ takes values corresponding to $1/(1-\sigma) \in \{1.34, 2.33, 3.27\}$, from Caunedo et al. (2023) at the lower endpoint, our baseline (2.33) in the middle, and the levels specification of Burstein et al. (2019) at the upper endpoint, and the within-occupation parameter σ_k takes values in $\{0, \text{baseline}, 2/3\}$, corresponding to within-occupation elasticities $1/(1-\sigma_k) \in \{1, \text{heterogeneous baseline}, 3\}$. The lower endpoint within occupations of $\sigma_k = 0$ is the Cobb–Douglas limit. Across the grid the aggregate output loss ranges from about 8 to 35 percent, and the variation follows the mechanism above. The largest losses arise in the Cobb–Douglas case ($\sigma_k = 0$), where majors are least substitutable and occupations that lack a field lose the most output; with the estimated within-occupation substitutability the loss falls substantially, and our baseline is near the lower end of the range. The loss is much less sensitive to the between-occupation elasticity: holding σ_k at its estimated values, moving σ from the baseline to either end of the range of the literature changes it by less than seven percentage points in every year bin. In each cell the productivity weights $\{\alpha^k_{j,t}\}$ and $\{\alpha_{k,t}\}$ are held at their baseline estimates, so the exercise asks how costly the observed allocation would be under the same technology with different substitutability.

Table 5: Aggregate Output Loss under Alternative Elasticities

cell	2003-2005	2006-2007	2008-2009	2010-2012	2013-2015	2016-2017	2018-2019
$\sigma = 0.254$, $\sigma_k = 0$ (Cobb-Douglas)	32.87	31.75	30.03	28.63	27.45	26.82	26.81
$\sigma = 0.694$, $\sigma_k = 0$ (Cobb-Douglas)	34.09	33.12	31.80	31.17	31.85	33.79	35.21
$\sigma = 0.571$, $\sigma_k = 0$ (Cobb-Douglas)	30.66	29.67	28.17	27.33	27.53	28.81	29.86
$\sigma = 0.254$, $\sigma_k = 2/3$ ($1/(1-\sigma_k) = 3$)	14.47	13.76	12.37	11.22	10.95	12.27	13.10
$\sigma = 0.694$, $\sigma_k = 2/3$ ($1/(1-\sigma_k) = 3$)	15.62	16.03	15.27	15.40	17.88	22.80	25.41
$\sigma = 0.571$, $\sigma_k = 2/3$ ($1/(1-\sigma_k) = 3$)	10.34	10.56	9.66	9.52	11.41	15.65	17.91
$\sigma = 0.254$, σ_k heterogeneous	15.63	14.71	13.08	11.77	11.31	12.25	12.91
$\sigma = 0.694$, σ_k heterogeneous	14.27	14.32	13.21	13.09	15.34	19.94	22.46
Baseline ($\sigma = 0.571$, σ_k heterogeneous)	9.89	9.77	8.50	8.10	9.73	13.55	15.67

Notes: Each cell reports the aggregate output loss (%) at equilibrium prices for the corresponding (σ, σ_k) pair, by year group. The baseline row is $\sigma = 0.571$ (i.e., $1/(1-\sigma) = 2.33$) with heterogeneous occupation-specific σ_k. The $\sigma_k = 0$ rows use the Cobb-Douglas limit branch of the equilibrium solver. Productivity weights are held at their baseline estimates in every cell.

6.3 Which Skills Drive Mismatch?

We next examine which of a major's skills account for the mismatch of its graduates, thereby identifying the origin of the wage losses documented above.[29] We treat each college major j as a bundle of L skills, $\boldsymbol{s}^j = (s_1^j, \ldots, s_L^j)$, priced in occupation k at skill prices $\{\psi_l^k\}$, and rewrite the earnings equation as

$$\ln w_{i,j,k,c,r,t} = \sum_{j'}^{J} \sum_{k'}^{K} \underbrace{\ln\Big(\sum_{l}^{L} \psi_l^{k'} s_l^{j'}\Big)}_{\beta_{j'}^{k'}} D_{i,j',k',c,r,t} + X_{i,t}\gamma + \lambda_{r,t} + \nu_c + e_{i,j,k,c,r,t}.^{30} \tag{29}$$

Skill mismatch for major j is the counterpart of Equation (8) with skill-priced values in place of returns,

$$\hat{m}_j^{micro} = \frac{\sum_{i\in\mathcal{I}^j}\Big(\sum_l^L \psi_l^{k^*(j)} s_{i,l}\Big)\tilde{h}_i - \sum_{i\in\mathcal{I}^j}\Big(\sum_l^L \psi_l^{k} s_{i,l}\Big)\tilde{h}_i}{\sum_{i\in\mathcal{I}^j}\Big(\sum_l^L \psi_l^{k} s_{i,l}\Big)\tilde{h}_i}, \tag{30}$$

where $s_{i,l}$ is skill l of worker i's 4-digit field of study, k is the worker's observed occupation, and $\tilde{h}_i$ the individual efficiency units of Section 2. Removing skill l from both terms of the numerator while holding the denominator fixed gives the contribution of skill l, $\sum_{i\in\mathcal{I}^j}(\psi_l^{k^*(j)} - \psi_l^k)s_{i,l}\tilde{h}_i$ divided by the same denominator, and the L contributions sum to $\hat{m}_j^{micro}$ by construction. For a major group, the measure and the contributions are the corresponding ratios of sums over the group's majors, as in Equation (9). We impute $\boldsymbol{s}^j$ for each major from ten O*NET knowledge classes, measured at the level of each worker's 4-digit field of study, so that skills vary across workers within a 2-digit major; Appendix B.4 describes the construction and the estimation of the skill prices, and decompose $\hat{m}_j^{micro}$ into the contributions of these ten skills, shutting down one skill at a time and recomputing mismatch; the resulting change in the mismatch level measures how much each skill drives (positive) or offsets (negative) mismatch. The ten contributions are additive, so each column of Table 6 sums to that group's baseline mismatch, reported in the last row, for the four major groups: STEM; Health and Education; Management and Commerce; and Society, Culture, and Arts.

Unlike the previous literature such as Deming (2017), we find no common skill factor, such as mathematics or general social skills, that accounts for wage losses across all majors. Instead, in three of the four groups the leading contributor is the knowledge class that defines the field. For STEM majors, *Engineering and Technology* skills account for about half of the group's mis-

[29] The construction of skills is described in Appendix B.4.

[30] In this decomposition, skills are unbundled in the sense that each skill is priced separately in each occupation. Edmond and Mongey (2021) study the case in which skills remain bundled within workers, so that wages price the bundle as a whole rather than its component skills; we do not consider that case here.

match (a contribution of 0.103 out of 0.200), with *Communications*, *Mathematics and Science*, and *Transportation* adding smaller amounts and the remaining classes offsetting. For Management and Commerce, *Business and Management* skills are the leading contributor (0.225 of 0.406), followed by *Communications*, *Education and Training*, and *Mathematics and Science*. For Society, Culture, and Arts, *Law and Public Safety* skills contribute more than the group's total (0.504 of 0.359), with *Business and Management* next and the science and health classes offsetting; law graduates, whose top occupation prices legal knowledge highly, drive this pattern. Health and Education is the exception: its mismatch is small (0.037), and the contributions are small and largely offsetting, with *Manufacturing and Production* (0.136) and *Health Services* (0.034) positive and most other classes negative. Economy-wide, the skill-priced measure is 22.9 percent, against 27.1 percent for the return-based measure of Table 3 on the same observations, so the ten skill prices reproduce most, but not all, of the return differences across occupations. In summary, mismatch arises when workers cannot deploy the field-specific knowledge that commands the highest return, which suggests that policies fostering general skills alone are unlikely to undo its costs. Two caveats apply. A field's skill vector is an occupation-weighted description of what its graduates' jobs require (Appendix B.4), so the decomposition does not separate knowledge taught by the field from knowledge acquired in the occupations its graduates hold. And because the ten class scores are highly correlated across fields, the individual contributions are estimated less precisely than the group totals.

Table 6: Skill-specific contributions to micro mismatch

	College Majors			
O*NET 10-class skill	STEM	Health and Education	Management and Commerce	Society, Culture, and Arts
Engineering and Technology	**0.103**	-0.036	-0.035	-0.066
Health Services	-0.019	0.034	-0.002	-0.073
Manufacturing and Production	-0.015	**0.136**	-0.030	-0.140
Law and Public Safety	-0.007	-0.045	0.057	**0.504**
Business and Management	-0.017	-0.005	**0.225**	0.121
Art and Humanities	-0.014	-0.019	0.001	-0.032
Communications	0.058	-0.002	-0.013	-0.018
Education and Training	0.014	-0.001	0.081	0.032
Mathematics and Science	0.052	-0.010	0.079	0.028
Transportation	0.045	-0.014	0.042	0.003
Total mismatch	0.200	0.037	0.406	0.359

Notes: Each cell reports the contribution of an O*NET 10-class skill to a major group's micro mismatch, Equation (30): the part of the skill-priced wage gain from moving to the top occupation $k^*(j)$ that is due to skill l, relative to the observed skill-priced wage bill. Positive values drive mismatch, negative values offset it; the contributions sum to the group's total in the last row. The largest contribution in each column is in bold. Skill prices are estimated by occupation on pooled person-year observations (Appendix B.4). Major groups: (1) STEM; (2) Health and Education; (3) Management and Commerce; (4) Society, Culture, and Arts. The economy-wide skill mismatch is 0.229; the micro mismatch of Table 3 computed on the same observations is 0.271.

7 Conclusion

This paper's contribution is a framework that measures skill mismatch consistently a micro, with fixed skill prices, and macro level, with variable skill prices. Defining mismatch as the deviation from the wage-return-maximizing allocation of college majors to occupations, rather than through task or qualification data, allows the same object to be evaluated under fixed returns, as an individual wage loss, and under equilibrium price adjustments, as an aggregate output loss. To illustrate the mechanism and the potential divergence between micro and macro estimates, we estimate the framework on Australian administrative tax panel data linking university degrees to longitudinal earnings and employment histories, correcting the wage returns for self-selection into occupations.

The comparison between the two measures yields the paper's main finding. The average individual wage loss from mismatch is 27.1 percent, but the corresponding output loss is 9.8 percent, roughly one-third, because prices adjust as workers reallocate, and graduates from different majors are, on our estimates, relatively substitutable within occupations. Estimates of mismatch that ignore these price adjustments therefore substantially overstate its aggregate cost. Mismatch is concentrated early in working life, it declines steeply with age and, once returns adjust, largely disappears by age 30. Commerce, Social Sciences, Arts, and STEM-related fields drive most of the aggregate loss. Policies concerned with the allocation of talent should accordingly focus on young workers, and consider their expected gains through a macro rather than micro lense.

Throughout, we have been deliberately agnostic about *why* workers end up mismatched. The non-pecuniary shocks in our Roy model are indistinguishably between search and informational frictions, licensing barriers, preferences, and unobserved amenities, and our measures quantify the cost of the resulting allocation regardless of its origin. This agnosticism is an advantage for measurement, but it limits the guidance our results offer for policy design. The ideal intervention depends on which friction generates the mismatch. Specifying and quantifying these causes is, in our view, the most important direction for future research.

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Appendix

A Additional Figures and Tables

Table A1: ASCED 2-digit College Majors

ASCED 2-digit	Major Title
1	Natural and Physical Sciences
2	Information Technology
3	Engineering and Related Technologies
4	Architecture and Building
5	Agriculture, Environmental and Related Studies
6	Health
7	Education
8	Management and Commerce
9	Society and Culture
10	Creative Arts

Notes: College majors used in the estimation sample, classified at the 2-digit ASCED level. Food, Hospitality and Personal Services (ASCED 11) is excluded from the estimation sample.

Table A2: ANZSCO 2-digit Occupations

ANZSCO 2-digit	Occupation Title
21	Arts and Media Professionals
22	Business, Human Resource and Marketing Professionals
23	Design, Engineering, Science and Transport Professionals
24	Education Professionals
25	Health Professionals
26	ICT Professionals
27	Legal, Social and Welfare Professionals
31	Engineering, ICT and Science Technicians
34	Electrotechnology and Telecommunications Trades Workers
36	Skilled Animal and Horticultural Workers
3X	Automotive, Engineering, and Construction Trades Workers
3Y	Food Trades, Other Technicians, and Trades Workers
41	Health and Welfare Support Workers
42	Carers and Aides
43	Hospitality Workers
45	Sports and Personal Service Workers
52	Personal Assistants and Secretaries
53	General Clerical Workers
54	Inquiry Clerks and Receptionists
55	Numerical Clerks
56	Clerical and Office Support Workers
59	Other Clerical and Administrative Workers
61	Sales Representatives and Agents
62	Sales Assistants and Salespersons
63	Sales Support Workers
7X	Machinery Operators and Drivers
8X	Labourers

Notes: Occupations used in the estimation sample, classified at the 2-digit ANZSCO level. Managerial and protective-service occupations are excluded from the sample (see Section 3). Codes ending in X or Y denote aggregated residual categories.

Figure A1: Major-Occupation Mismatch Estimates

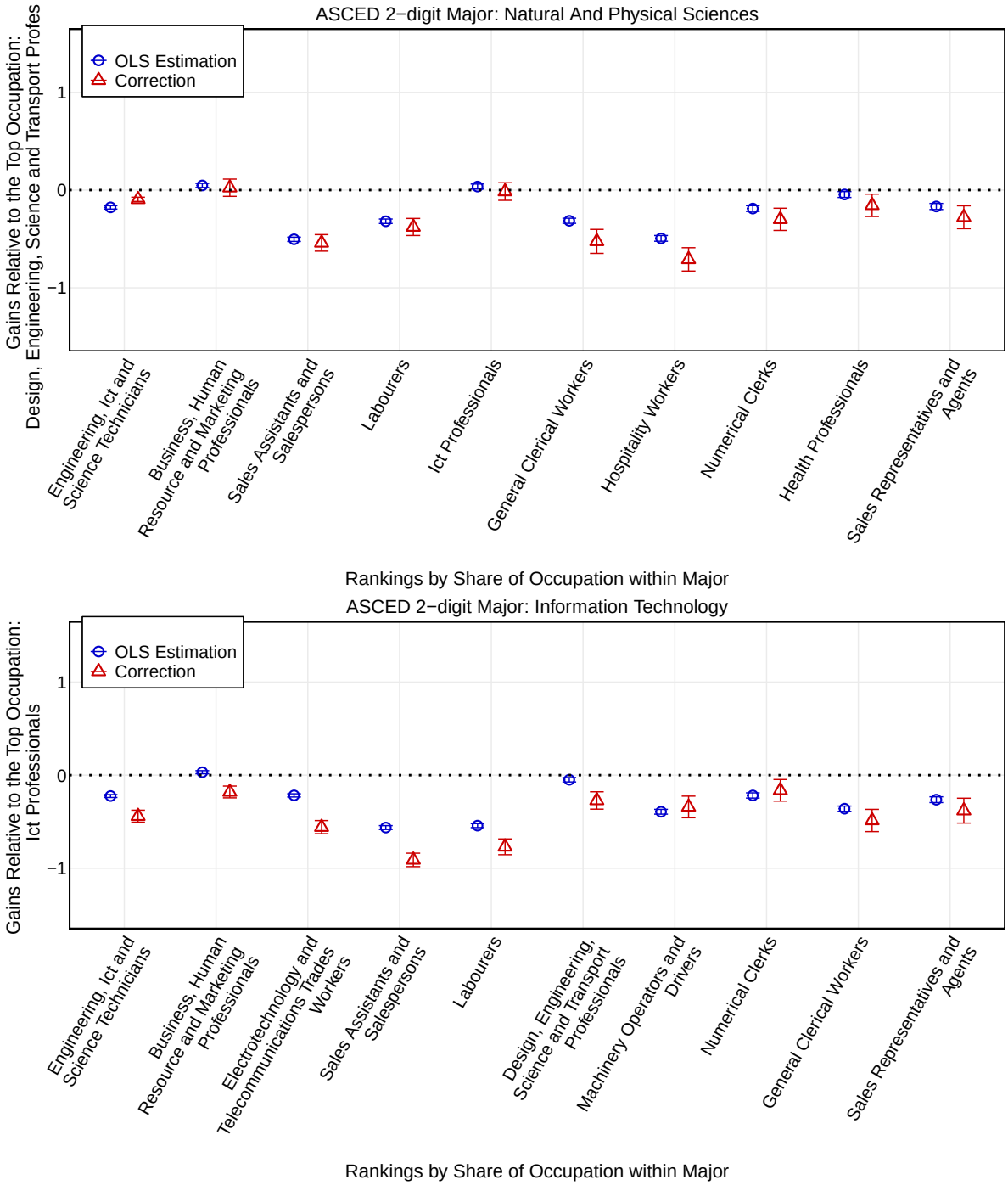


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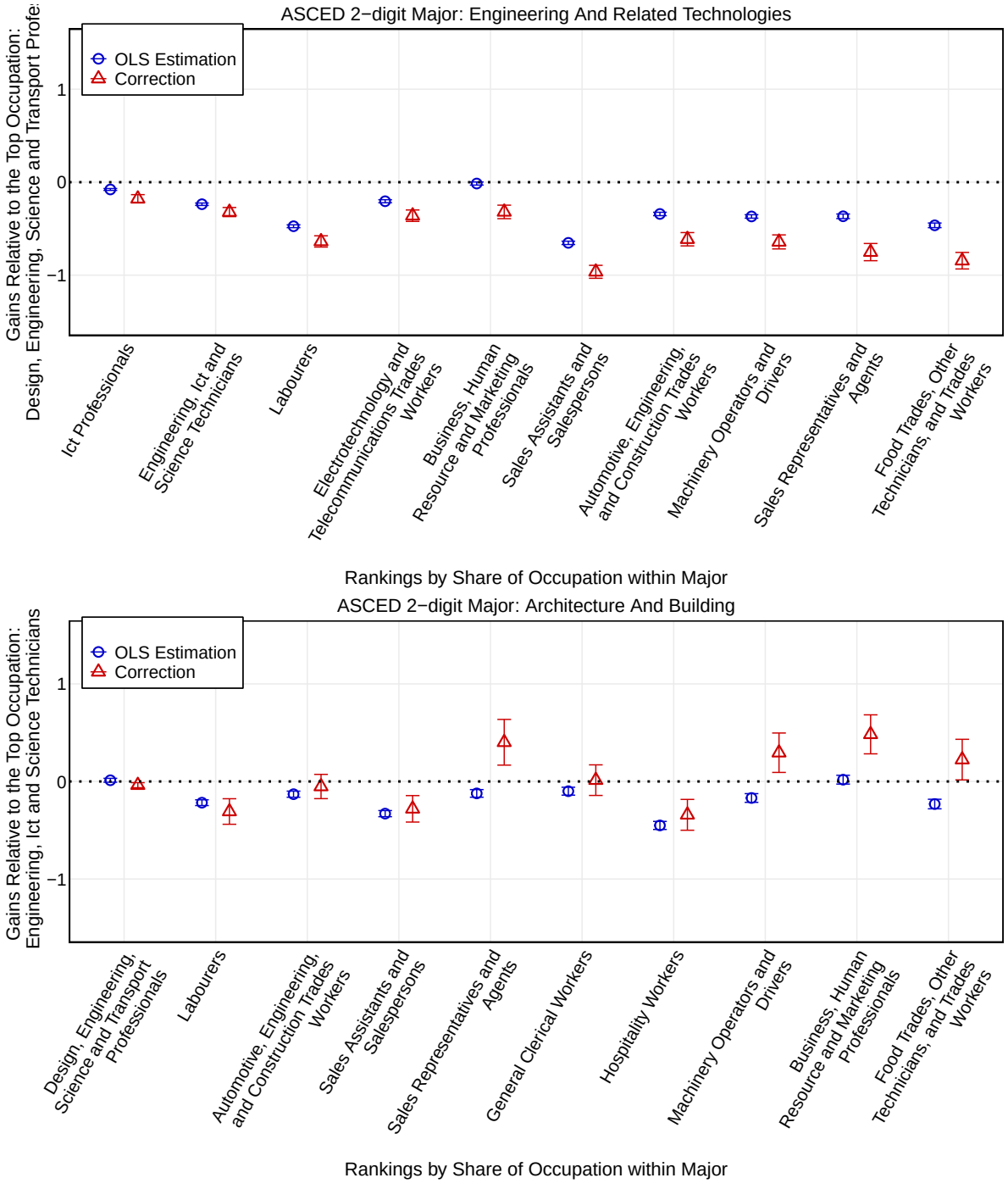


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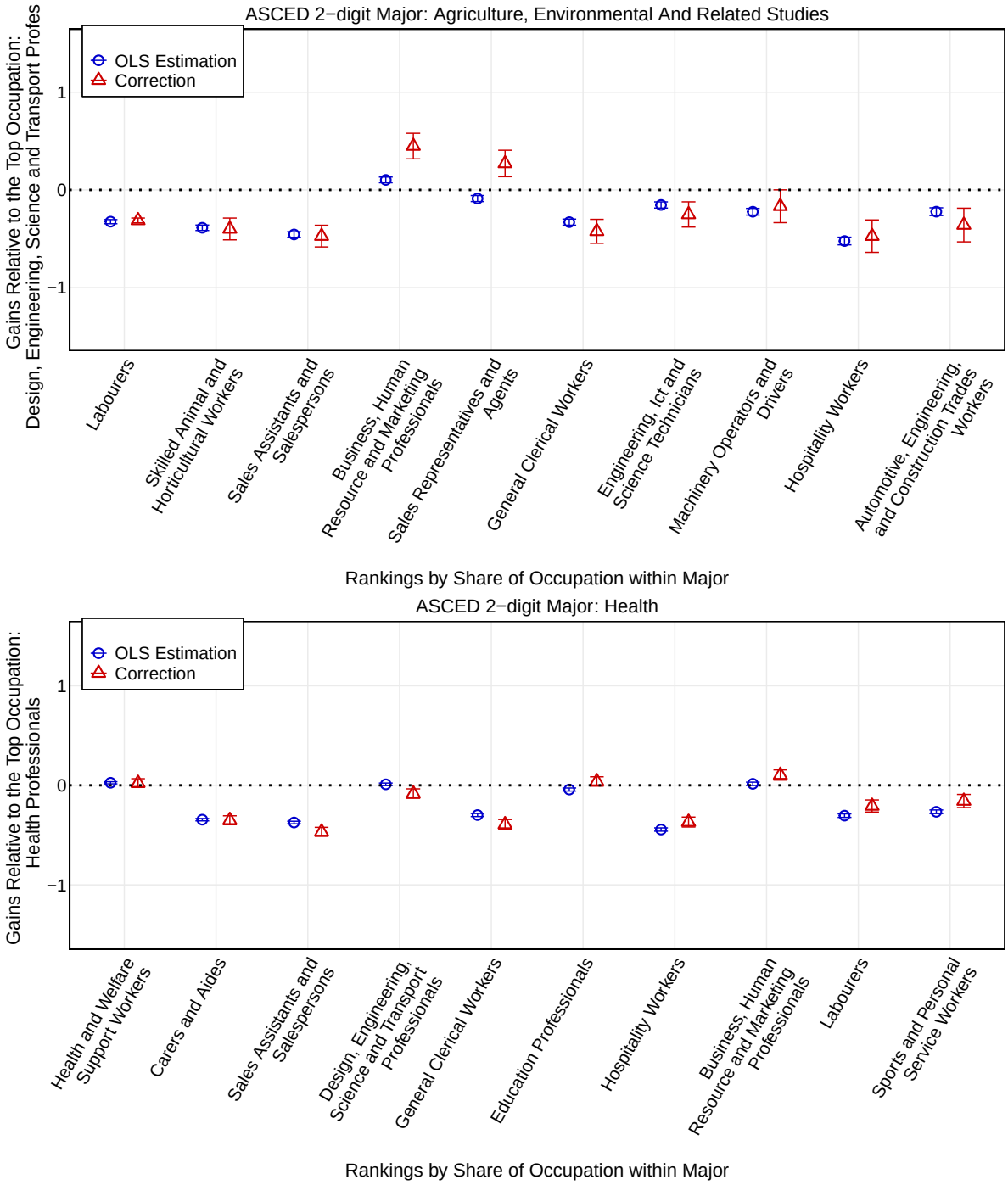


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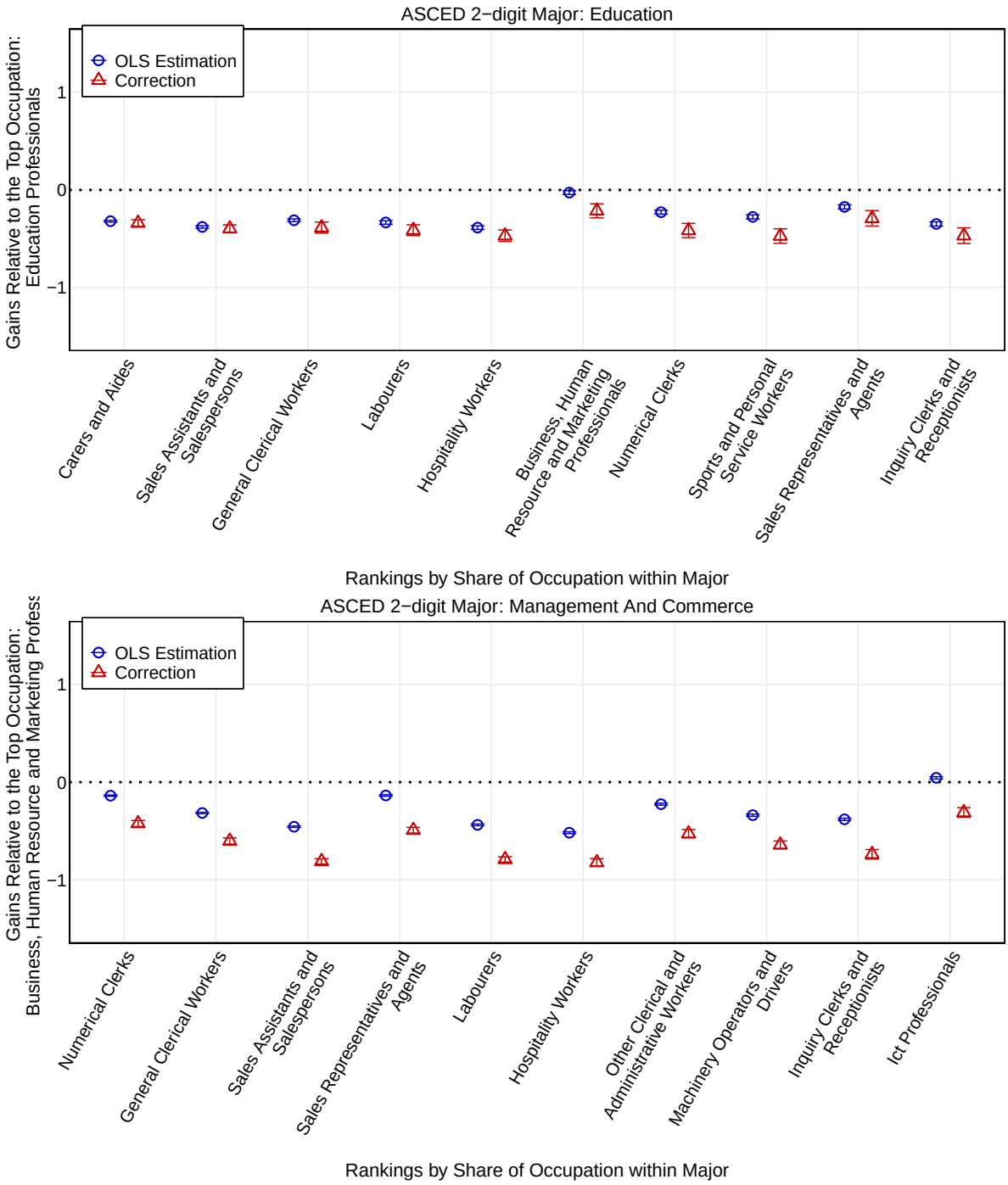
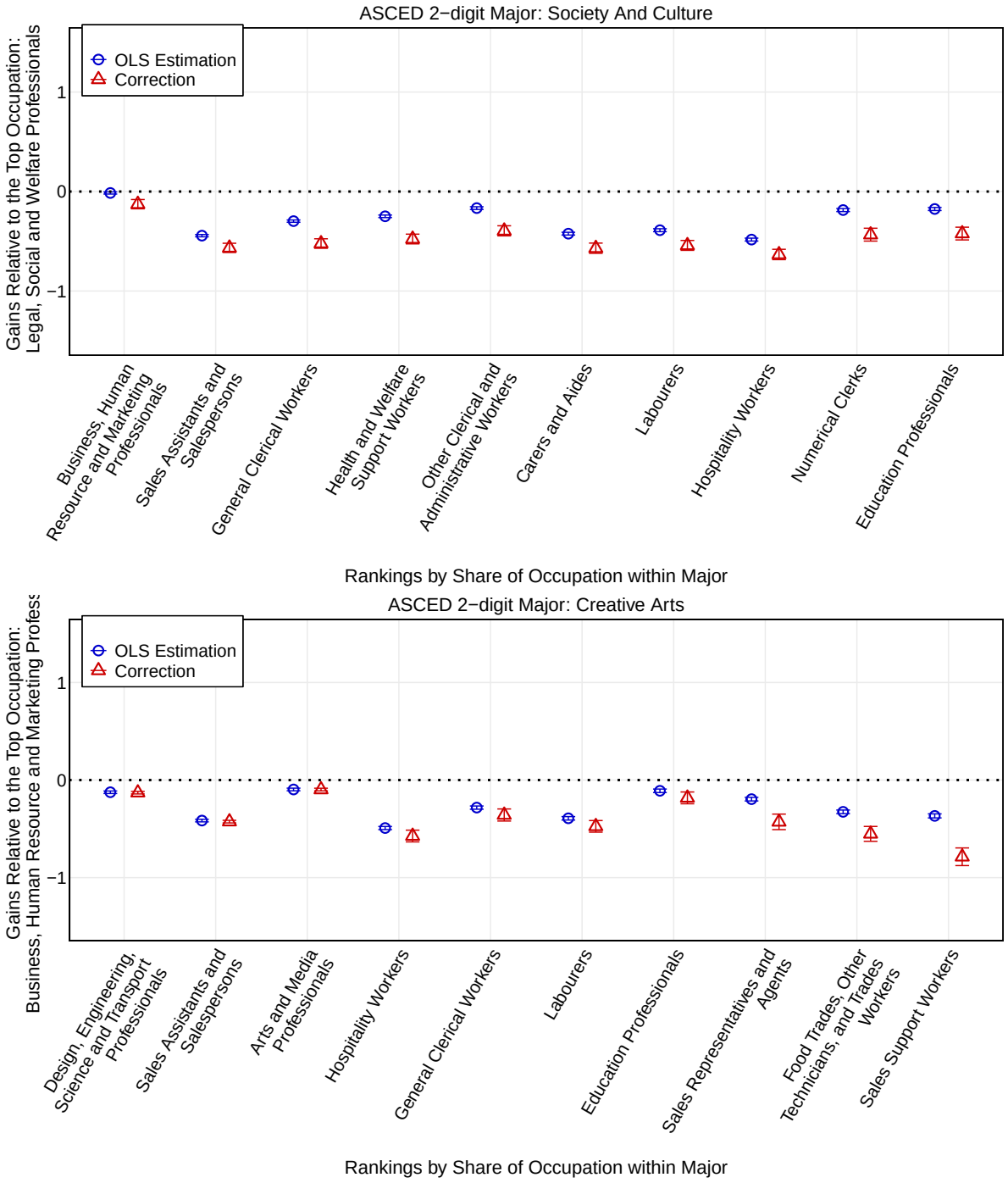
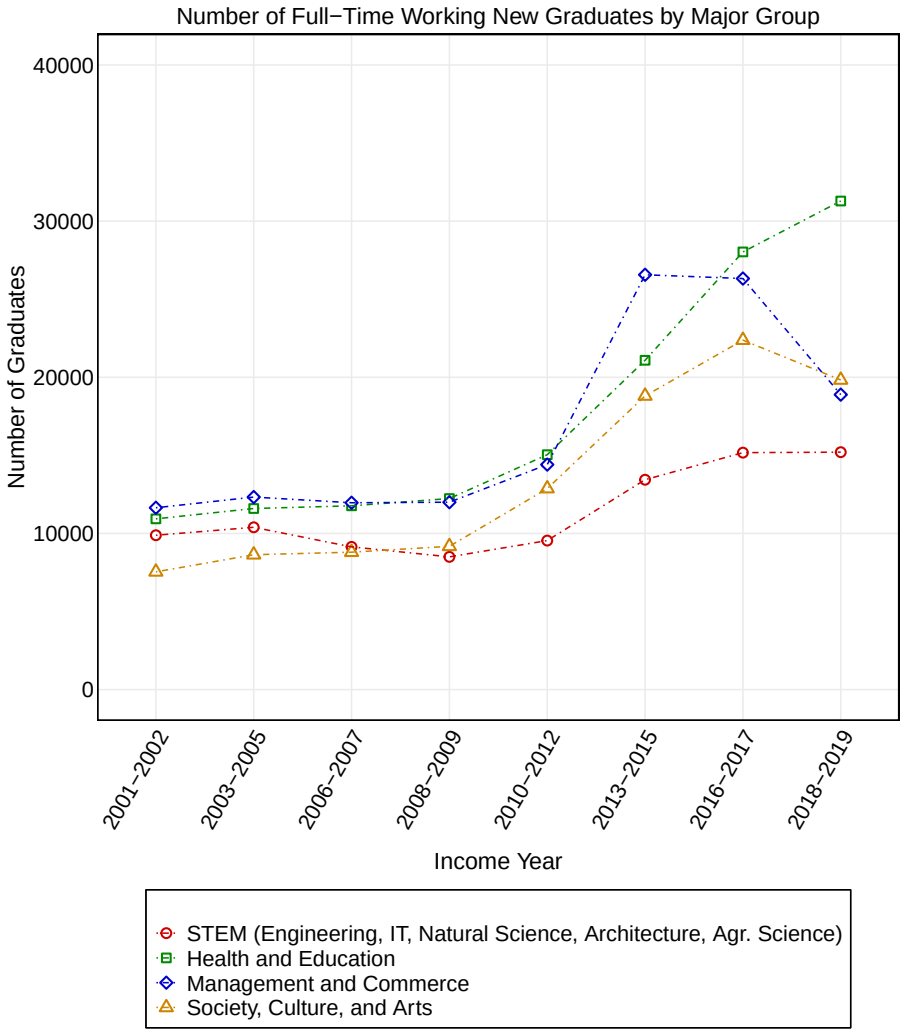


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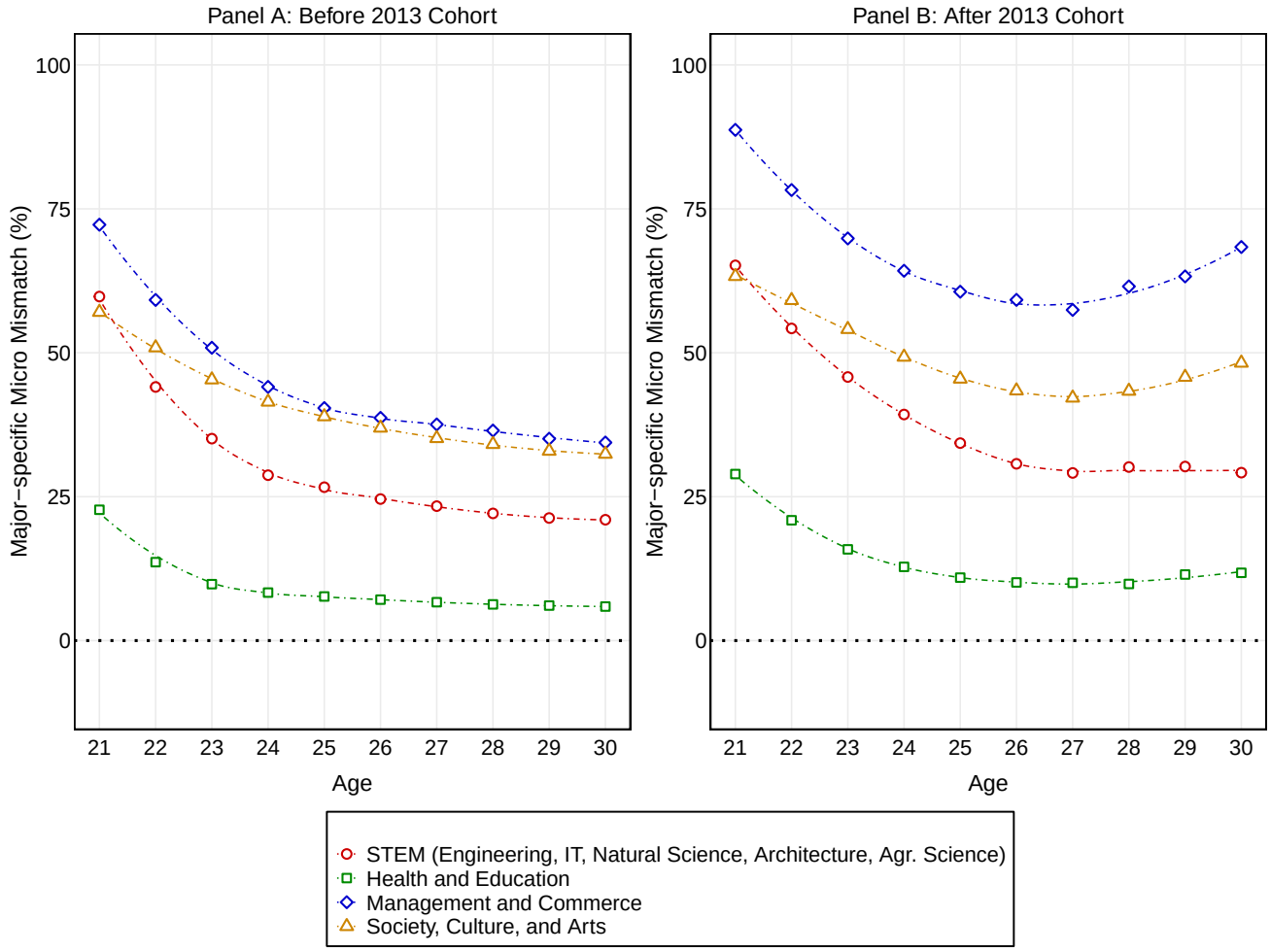
Notes: These figures plot the wage penalties from not choosing the field top-occupation (as shown in y-axis) at each observed occupational choice (as shown in x-axis) for each field of study. The order of occupations in the x-axis is based on the descending order of the share of employment, ordered from Left to Right. The scale of y-axis represents the relative gain (or loss) in log earnings from not choosing the top-occupation. Both the OLS (blue circle) and corrected point (red triangle) estimate of the wage penalties $\zeta_{j,k}$ from Equation (20) and 90% confidence interval are shown. To avoid cluttering the figure, only reports the top 10 occupations, by employment shares, within the major. The top occupation is the highest-return occupation among those employing at least 10 percent of the major's graduates (Section 3), so an occupation with a smaller share can show a positive value: its return exceeds that of the top occupation, but it is excluded from the benchmark.

Figure A2: Number of New (Full-time Working) Graduates By Majors



Notes: This figure plots the number of newly full-time working graduates by each major over time.

Figure A3: The Average Individual Mismatch over the Life-Cycle by Cohorts



Notes: This figure plots the evolution of occupational mismatch for aggregate fields of study over the life-cycle. Cohorts are split pre- and post-policy intervention in 2009, such that cohorts graduating from 2013 onward are assumed to have studied under an uncapped system of Commonwealth-supported places. The estimate of field-specific occupational mismatch is computed using Equation (8) for each age by each cohort group. Because the data end in 2019, graduates from 2013 onward observed at older ages completed their degree late (at age 24 or older for an observation at age 30), so the two cohorts have comparable composition only at the younger ages.

Figure A4: Occupational Distribution by Age: Data versus Aggregate (Macro) Optimum

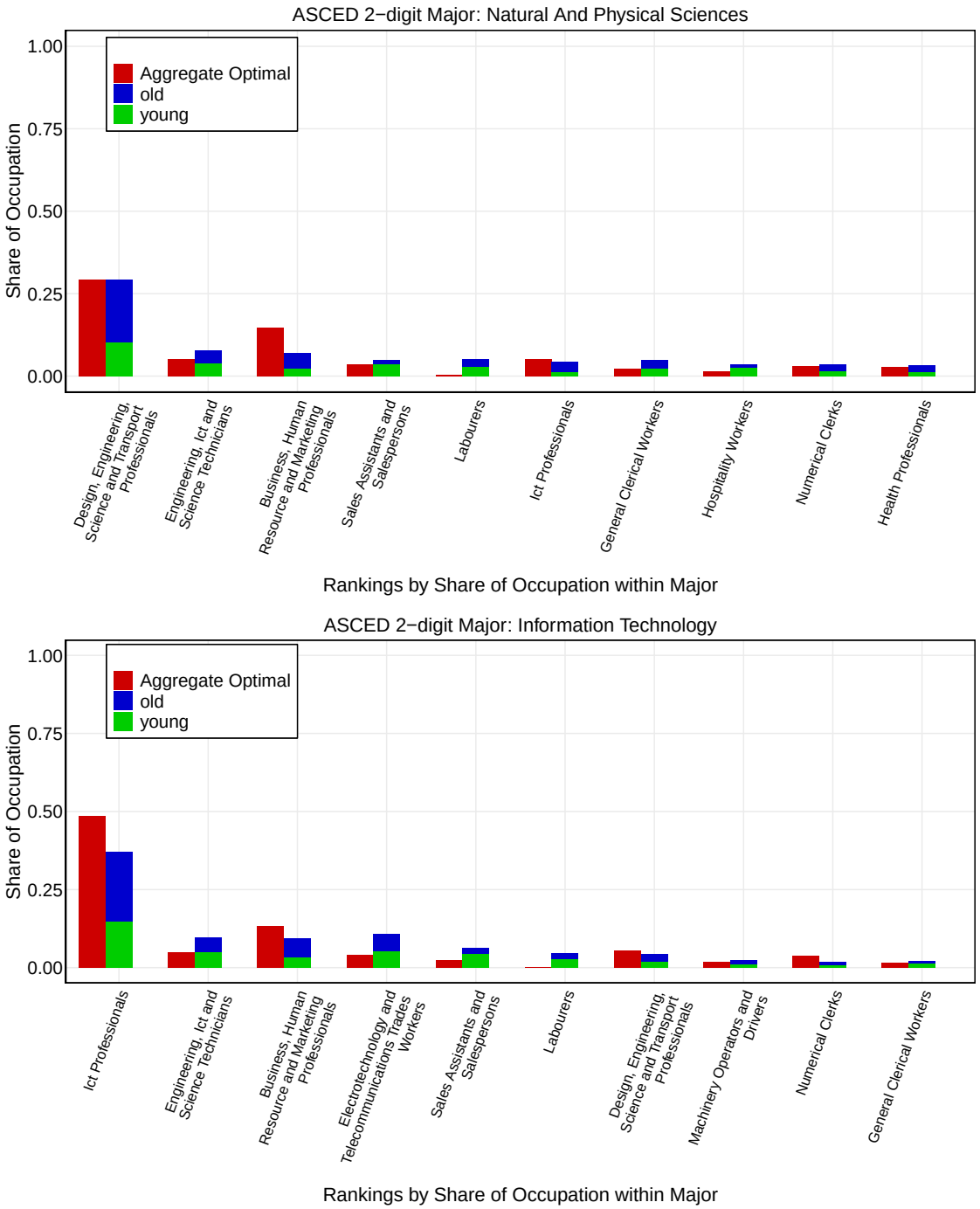
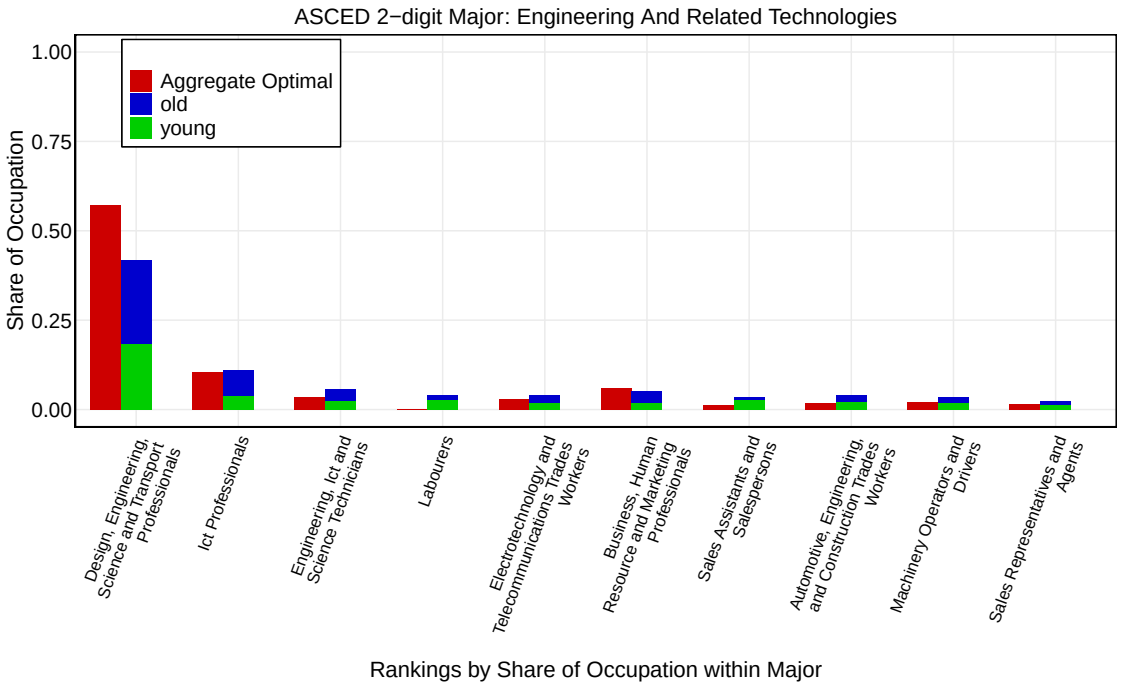


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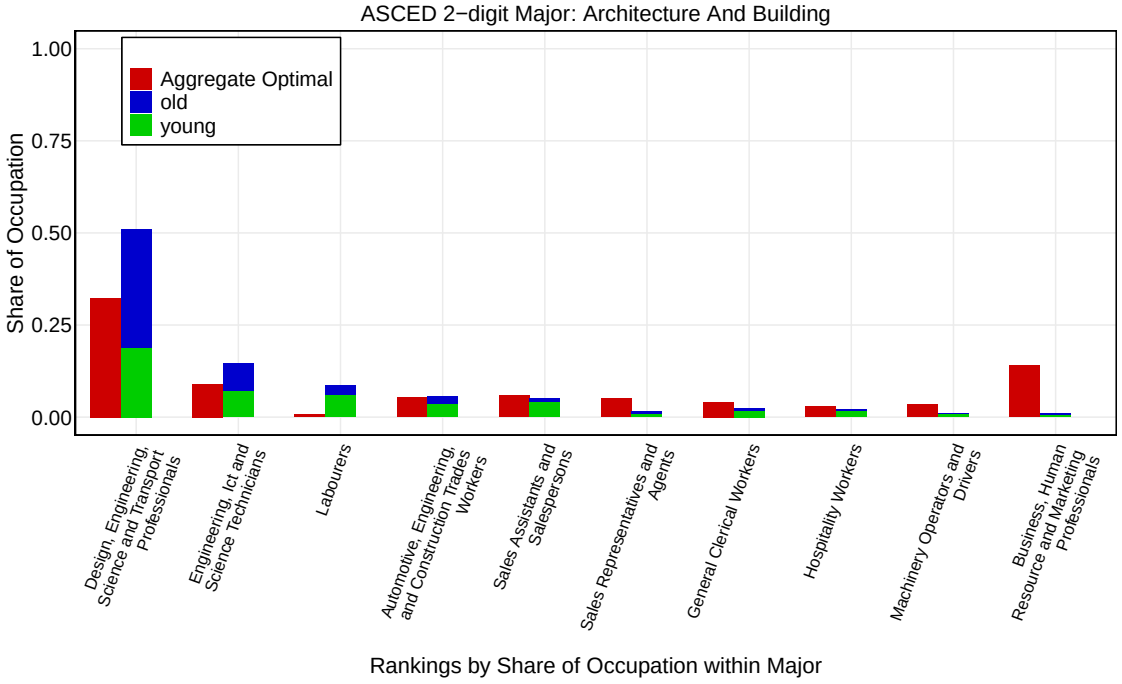
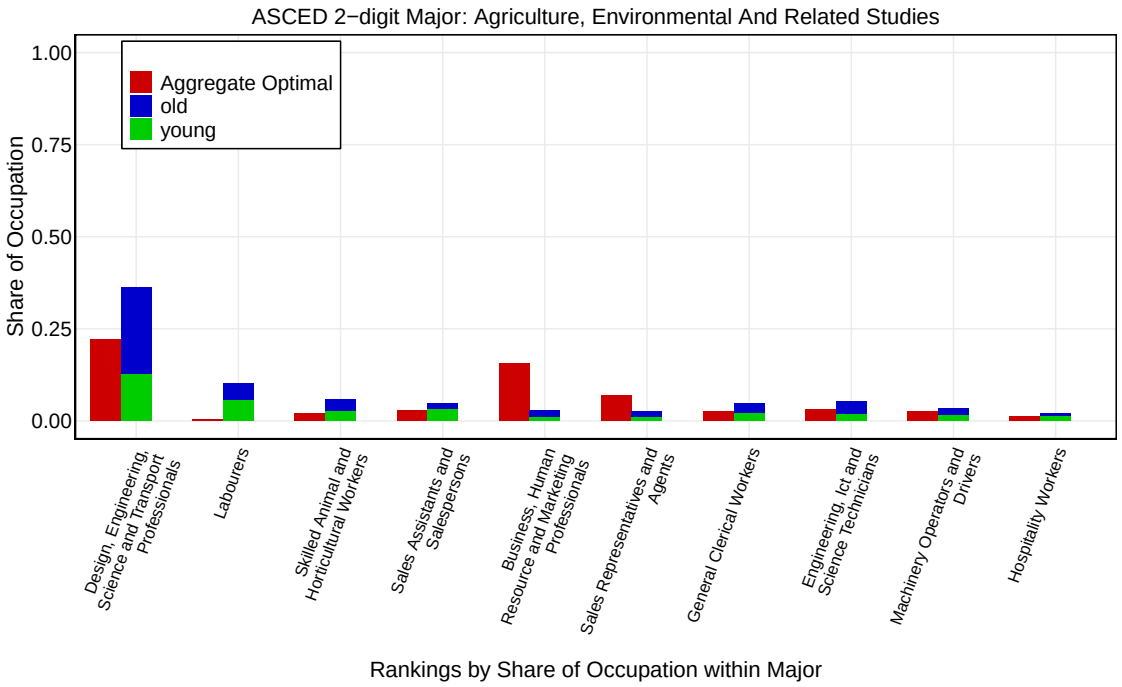


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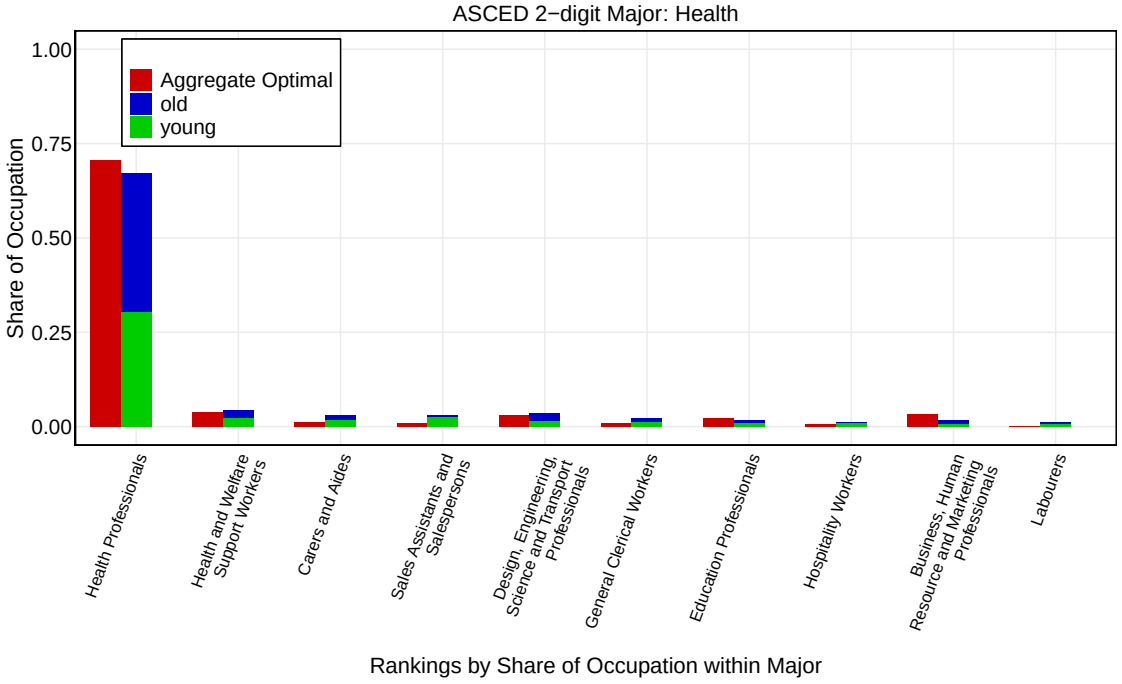
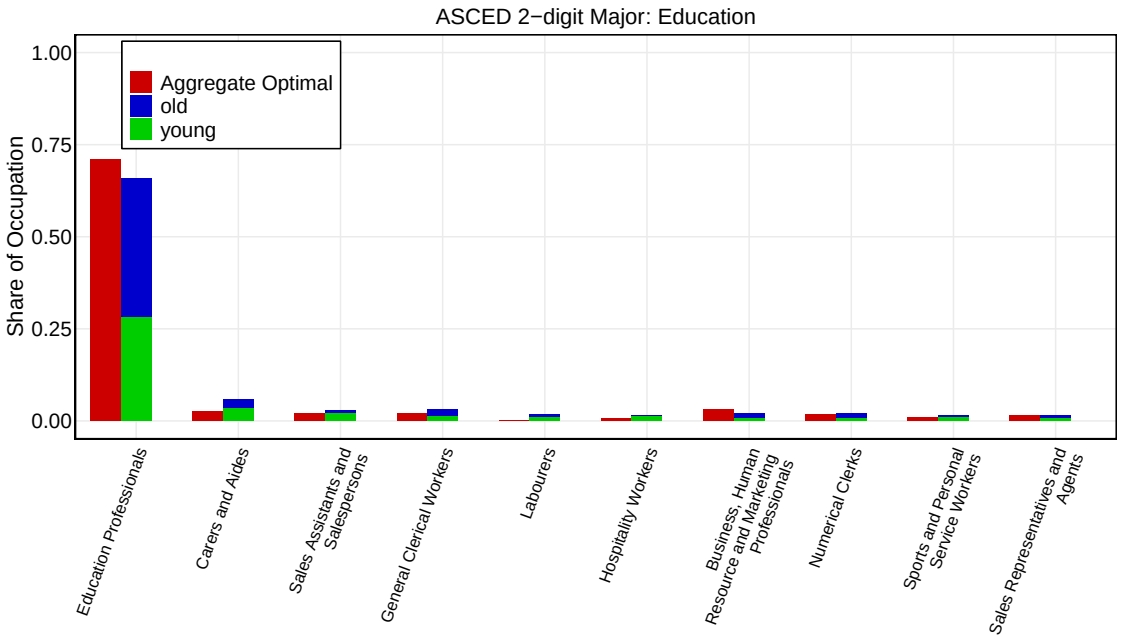


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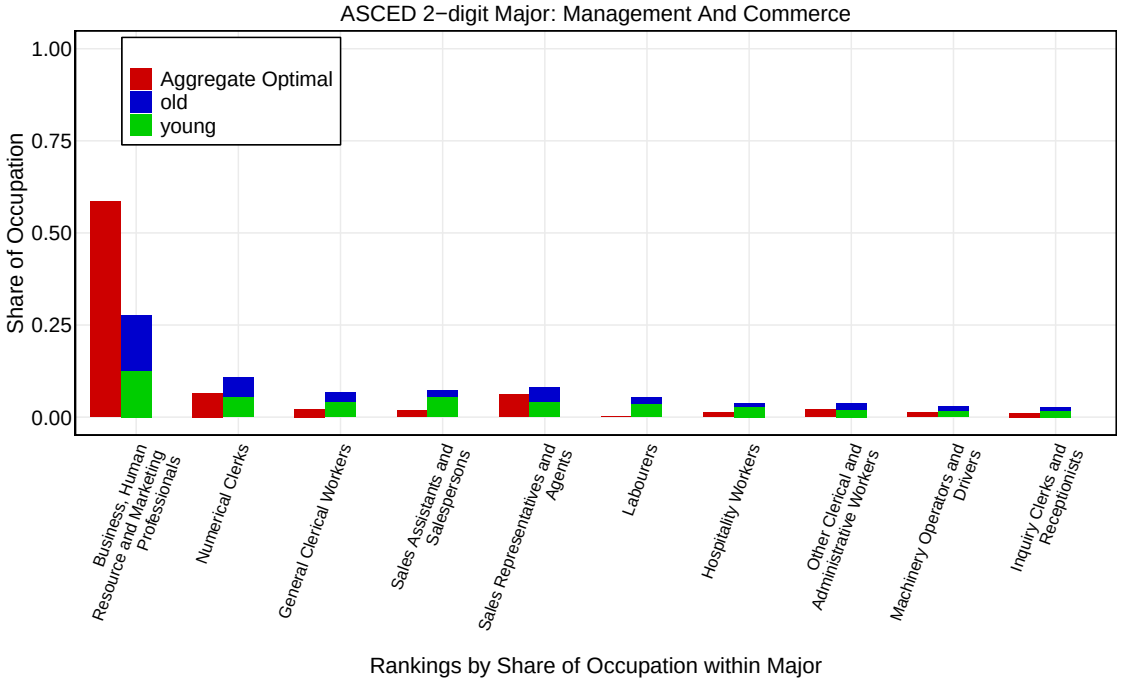
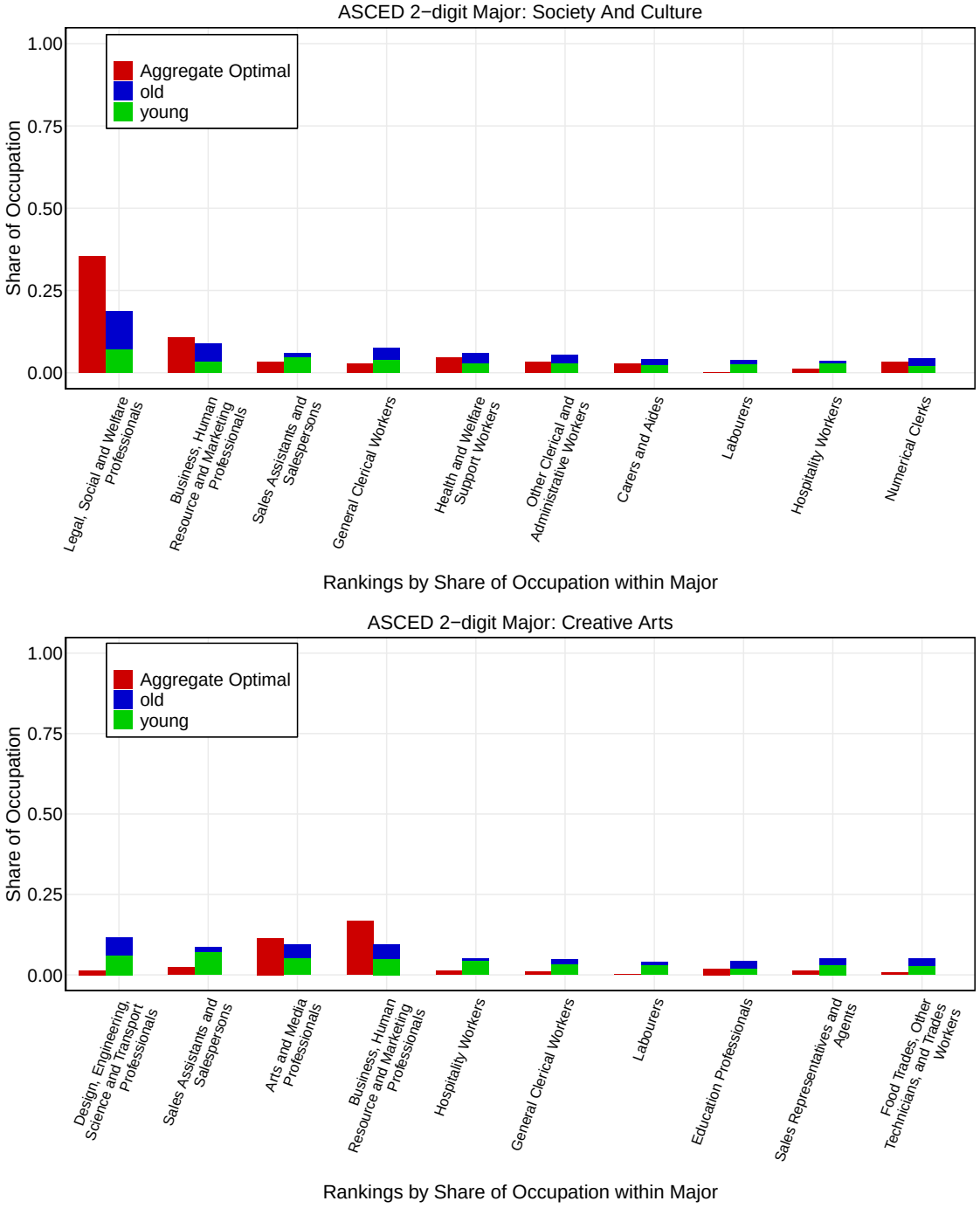


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Notes: These figures plot, for each field of study, the observed occupational distribution of young workers (aged 21–28) and old workers (aged 29–35), stacked, against the efficient aggregate allocation of that field across occupations. Depicted are shares of labor efficiency units within each major, so the stacked bar and the aggregate-optimal bar each sum to one across all occupations of the major. The order of occupations in the x-axis is based on the descending order of the observed share of the occupation within the major, ordered from left to right. To avoid cluttering the figure, we only show the shares of the occupations whose occupational share is among the top 10 of all occupations within the major.

B Data

B.1 HECS/HELP Data

We use Higher Education Loan Program (HELP) data to obtain an individual's post-secondary educational information. This data provides us with the worker's college course name (i.e., Bachelor of Engineering) which suggests both degree level and field of study; institutional types such as university, Technical and Further Education (TAFE)/College/Academy, or school/others; and yearly student loan debt. We design a keyword extraction algorithm to classify thousands of different course names recorded in HELP into the 4-digit level of the Australian Standard Classification of Education (ASCED) as well as into different degree levels.[31] Since degree completion indicators are not available in the administrative data, we follow the approach by Andrews, Deutscher, Hambur, and Hansell (2020) and impute the completion indicator by observing when students stop incurring new HECS/HELP debt for each degree level. As such, the graduation cohort is defined as the last year when the worker incurs the new debt.

B.2 Australian Longitudinal File on Individuals (ALife)

ALife tax records provide us with annual pre-tax salary and wage data, the occupational track record (classified by ANZSCO codes), and place of residence for 10% of the Australian population. We aggregate workers' occupations at the ANZSCO 2-digit level. We construct an individual's work experience as the number of years, up to and including the current one, in which the worker is observed with a non-missing occupation and annual earnings at or above the earnings floor; years below the floor are not counted.[32] We inflate wages to 2019 Australian dollars.

B.3 Estimation Sample and Restriction

We link workers' labor market outcomes from ALife to their college major and restrict observations to the period after the year of graduation. To ensure that we have sufficient samples within each major-occupation pair, we use 2-digit levels of ASCED and ANZSCO codes to classify majors and occupations, respectively. We focus on the undergraduate sample, which is limited to individuals between the ages of 21-35 years old who held at least one type of

[31] Specifically, degree level is aggregated based on ASCED into three broad groups: (1) non-award degree; (2) certificate/diploma and bachelor; and (3) graduate certificate/diploma and postgraduate degree.

[32] The floor is half of full-time, full-year earnings at the national minimum wage, floor = minimum hourly wage/2×35 hours $\times$ 48 weeks, as stated in Section 3.

academic degree, such as a diploma, certificate, or bachelor's degree, regardless of institution type; person-years in which the most recently completed qualification is a postgraduate degree, a graduate certificate or diploma, or a non-award course are excluded. To ensure that wages reflects the skill prices and to alleviate the concern of the impact of career promotion on wages, we remove observations that are self-employed or work in managerial (ANZSCO major group 1), protective-service (sub-major group 44), or office-manager and program-administrator (sub-major group 51) positions. It is not uncommon to find a worker who has studied two different majors, and/or obtained more than one degree level throughout their higher education history. To ensure that occupations are matched to only one field of study and degree level, we restrict the sample to individuals with only one major at the 2-digit classification and the most recent degree level.

B.4 Skill Data Construction

The worker's skill data is constructed at the ASCED 4-digit level to provide us with more variation in estimating the skill price. To construct precise skill data for each ASCED 4-digit field of study, we start from the most granular level of major/occupation code available and utilize a couple of standard crosswalks to impute the skills data for each major from the skills data that we computed at the ANZSCO 3-digit occupation level. We proceed with the imputation in four main steps:

1. *Construct Skills Data at ANZSCO 3-digit Occupation Level.* The construction is done by imputing the skill data on O*NET occupation to ANZSCO 3-digit occupation based on several crosswalks: (1) *ANZSCO to ISCO-08*; (2) *ISCO-08 to SOC10*; and (3) *SOC10 to O*NET SOC*. With these, we construct a crosswalk of *ANZSCO to O*NET SOC*. O*NET rates each occupation on 33 knowledge items.[33] We impute the set of O*NET skill data to ANZSCO 3-digit level. To this end, all ANZSCO 3-digit codes have a set of parallel skill data as O*NET SOC codes.

2. *Link ASCED to ANZSCO through Crosswalk CIP-SOC10.* The core in this step is the crosswalk *CIP-SOC10*. This crosswalk provided by the US National Center for Education Statistics matches the post-secondary fields of study providing graduates with specific skills and knowledge to the occupations requiring those skills or knowledge to be successful.[34]. The *CIP-SOC10* is U.S. based. However, it allows us to construct a parallel crosswalk from ASCED to ANZSCO by (1) mapping ASCED to CIP; and (2) mapping

[33]More details can be found at https://www.onetonline.org/find/descriptor/browse/2.C
[34]More detail can be found at https://nces.ed.gov/ipeds/cipcode/post3.aspx?y=56

ANZSCO to SOC (which is completed in step 1). We execute step (1) by using existing *Crosswalk CIP to ISCED* and *Crosswalk ISCED to ASCED*. To this end, we construct a major-occupation crosswalk ASCED to ANZSCO, which is Australia based.

3. *Impute Skill from ANZSCO to ASCED through Differential Weights*. One issue from our constructed crosswalk ASCED to ANZSCO is that two different 4-digit ASCED codes may map into the same set of 3-digit ANZSCO codes, which would reduce the variation in major skills if simply imputing the skills to major by taking the average. We address this by utilizing different weights of 3-digit ANZSCO occupation within each 4-digit ASCED major when executing our imputation, with weights as the number of individuals in each 3-digit ANZSCO occupation within each 4-digit ASCED major. Namely, the skills of occupation with higher weights are weighted more in imputing the skills of matched major. To this end, for each 4-digit ASCED major, we have a set of parallel 33 knowledge items from O*NET.

4. *Reduce the Dimensionality of the Imputed Major Skills.* We further reduce the dimension of major skills to have a meaningful interpretation. We aggregate the 33 knowledge items into the ten knowledge classes of O*NET's Knowledge taxonomy (2.C.1–2.C.10), taking the first principal component of the items in each class, and rescale each class score to the unit interval by a linear transformation, so that the distance between two 4-digit ASCED majors is comparable across classes. This yields ten skills for each 4-digit ASCED major.

Estimation of skill prices. We estimate the skill prices $\{\psi_l^k\}$ in Equation (29) separately for each occupation k. The dependent variable is the person-year's log earnings net of the controls in Equation (20), that is, the estimated major–occupation return plus the residual. The skill vector $\boldsymbol{s}_i$ is that of the worker's 4-digit ASCED field of study, averaged across fields for workers with more than one and scaled by 100. For each occupation, we fit $\ln\left(\sum_l \psi_l^k s_{il}\right)$ to this dependent variable by nonlinear least squares on the pooled person-year observations in that occupation, without an intercept and subject to $\psi_l^k \geq 0$. No further normalization is imposed: without an intercept, the level of $\{\psi_l^k\}$ absorbs the average earnings level of occupation k. Skill prices are constant over time. Equation (30) then values each person-year at its own skill vector.

C Estimation of the Wage Equation

C.1 Correct Selection Bias

Individuals are observed in an j,k cell if and only if the Rule (19), conditional on major j, is satisfied,[35]

$$M_{i,j,k,c,r,t} = 1, \text{ iff } e_{i,j,k',c,r,t} - e_{i,j,k,c,r,t} \leqslant \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t};\ \forall k' \neq k \tag{C.1}$$

Directly using (C.1) to pin down the selectivity bias is infeasible because we would need a complete specification of the joint distribution of error terms in the wage equation and error terms in K selection equations for each major j. To address the estimation challenge, we follow Lee (1983) to reduce the problem by only looking at the maximum order statistic. Equation (C.1) can be rewritten as,

$$M_{i,j,k,c,r,t} = 1, \text{ iff } \max_{k'}(e_{i,j,k',c,r,t} - e_{i,j,k,c,r,t} - (\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t})) \leqslant 0 \tag{C.2}$$

Given this, the observation in cell (j,k) is observed ($M_{i,j,k,c,r,t} = 1$) if the maximum order statistics is less than 0. To get the probability of this event, let $F^e_{k|j}$ be the joint CDF of selection error terms $(e_{i,j,1,c,r,t} - e_{i,j,k,c,r,t}, ..., e_{i,j,K,c,r,t} - e_{i,j,k,c,r,t})$. We derive the CDF of the maximum order statistics, conditional on sub-utility differences,

$$\begin{aligned}
&H_{k|j}(t \mid \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t}, ..., \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,K,c,r,t}) \\
=&Pr(\max_{k'}(e_{i,j,k',c,r,t} - e_{i,j,k,c,r,t} - (\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t})) \leqslant t \mid \\
&\quad \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t}, ..., \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,K,c,r,t}) \\
=&Pr(e_{i,j,1,c,r,t} - e_{i,j,k,c,r,t} \leqslant \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t} + t, ..., \\
&\quad e_{i,j,K,c,r,t} - e_{i,j,k,c,r,t} \leqslant \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,K,c,r,t} + t) \\
=&F^e_{j,k}(\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t} + t, ..., \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,K,c,r,t} + t)
\end{aligned} \tag{C.3}$$

[35] Note, we allow for regional variation on selection probabilities giving us additional variation in the data.

If $t = 0$, then

$$\begin{aligned} &Pr(\max_{k'}(e_{i,j,k',c,r,t} - e_{i,j,k,c,r,t} - (\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t})) \leqslant 0 \\ &\quad |\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t}, ..., \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,K,c,r,t}) \\ =&F^e_{j,k}(\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t}, ..., \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,K,c,r,t}) \\ =&F^e_{j,k}(\vec{U}) \end{aligned} \tag{C.4}$$

where the vector of sub-utility difference is $\vec{U} = (\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t}, ..., \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,K,c,r,t})$.

To this end, Equation (C.4) gives the probability that the maximum order statistics $\max_{k'}(e_{i,j,k',c,r,t} - e_{i,j,k,c,r,t} - (\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t}))$ is less than 0. That is the probability of observing an individual in occupation k conditional on major j. However, these random variables indexed over i are not identically distributed because the sub-utility differences might be systematically different.

Following Lee (1983), we propose a new random variable that can be constructed using the transformation,

$$v^*_{i,j,c,k,r,t} = \Gamma^{-1}_{j,k}\{F^e_{j,k}(\vec{U})\}, \tag{C.5}$$

where $\Gamma^{-1}_{j,k}$ can be any univariate continuous strictly increasing function, so that $\Gamma_{j,k}$ can be defined as a marginal distribution of the new random variable $v_{i,j,k,c,r,t}$.[36] Concretely, let $\varepsilon^{\max}_{i,j,k,c,r,t} = \max_{k'}(e_{i,j,k',c,r,t} - e_{i,j,k,c,r,t} - (\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t}))$ denote the realized maximum order statistic and $H_{k|j}(\cdot \mid \vec{U})$ its conditional distribution function from Equation (C.3), and define $v_{i,j,k,c,r,t} = \Gamma^{-1}_{j,k}\{H_{k|j}(\varepsilon^{\max}_{i,j,k,c,r,t} \mid \vec{U})\}$. This is a strictly increasing transform of the realized statistic whose marginal distribution is $\Gamma_{j,k}$ by construction, so $\varepsilon^{\max}_{i,j,k,c,r,t} \leqslant 0$ if and only if $v_{i,j,k,c,r,t} \leqslant \Gamma^{-1}_{j,k}\{H_{k|j}(0 \mid \vec{U})\} = v^*_{i,j,k,c,r,t}$. Therefore, the probability of a random variable $v_{i,j,k,c,r,t}$ that is less or equal to the value $v^*_{i,j,k,c,r,t}$ can summarize the probability of the event that the maximum order statistics is less than 0, and thereby $M_{i,j,k,c,r,t} = 1$ and,

$$Pr(v_{i,j,k,c,r,t} \leqslant v^*_{i,j,k,c,r,t}) = \Gamma_{j,k}(\Gamma^{-1}_{j,k}\{F^e_{j,k}(\vec{U})\}) = F^e_{j,k}(\vec{U}). \tag{C.6}$$

Hence, using (C.5), we can rewrite (C.2) as the follows:

$$M_{i,j,k,c,r,t} = 1, \text{ iff } v_{i,j,k,c,r,t} \leqslant \Gamma^{-1}_{j,k}\{F^e_{j,k}(\vec{U})\}. \tag{C.7}$$

Following Lee (1983), we assume that the error in wage equation $\epsilon_{i,j,k,c,r,t}$ and this new transformed random variable $v_{i,j,k,c,r,t}$ are independently and identically distributed with a joint

[36] The implicit assumption in Lee (1983) for the above transformation is that the same transformation is applied regardless of the specific values of sub-utility differences.

distribution function $G_{j,k}$. With this assumption, we can use a two-stage Heckman estimation for correcting selectivity bias. Specifically, we assume that: (1) $\Gamma_{j,k}$ is a univariate standard normal CDF (Φ); (2) $G_{j,k}$ is a bivariate standard normal CDF.

To this end, we pin down the self-selection bias $E[\epsilon_{i,j,k,c,r,t}|M_{i,j,k,c,r,t}=1]$ based on the above distributional assumption in the following way:

$$
\begin{aligned}
&E[\epsilon_{i,j,k,c,r,t}|M_{i,j,k,c,r,t}=1] \\
&\quad = E[\epsilon_{i,j,k,c,r,t}|v_{i,j,k,c,r,t} \leqslant \Phi_{j,k}^{-1}\{F_{j,k}^{e}(\vec{U})\}] \\
&\quad = \frac{Cov(\epsilon_{i,j,k,c,r,t}, v_{i,j,k,c,r,t})}{var(v_{i,j,k,c,r,t})} E(v_{i,j,k,c,r,t}|v_{i,j,k,c,r,t} \leqslant \Phi_{jk}^{-1}\{F_{j,k}^{e}(\vec{U})\}) \\
&\quad = \rho_{j,k}\,\sigma_{\epsilon,j,k}\,E(v_{i,j,k,c,r,t}|v_{i,j,k,c,r,t} \leqslant \Phi_{j,k}^{-1}\{F_{j,k}^{e}(\vec{U})\}) \\
&\quad = \rho_{j,k}\,\sigma_{\epsilon,j,k}\frac{-\phi(\Phi^{-1}(F_{j,k}^{e}(\vec{U})))}{\Phi(\Phi^{-1}(F_{j,k}^{e}(\vec{U})))} \\
&\quad = \rho_{j,k}\,\sigma_{\epsilon,j,k}\frac{-\phi(\Phi^{-1}(F_{j,k}^{e}(\vec{U})))}{F_{j,k}^{e}(\vec{U})} \\
&\quad = \rho_{j,k}\,\sigma_{\epsilon,j,k}\lambda_{j,k}(F_{j,k}^{e}(\vec{U}))
\end{aligned}
\tag{C.8}
$$

where $\vec{U} = (\tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,1,c,r,t}, ..., \tilde{U}_{i,j,k,c,r,t} - \tilde{U}_{i,j,k',c,r,t})$, $\rho_{j,k}$ is the correlation between $\epsilon_{i,j,k,c,r,t}$ and $v_{i,j,k,c,r,t}$, $\sigma_{\epsilon,j,k}$ is the standard deviation of $\epsilon_{i,j,k,c,r,t}$ (the variance of v is one by construction), so the coefficient on the correction function is $\rho_{j,k}\sigma_{\epsilon,j,k}$, which we estimate without restriction, and $\lambda_{j,k}(\cdot)$ is the parametric form of the correction function.[37] Following Dahl (2002) with the index sufficiency assumption, we can show that $F_{j,k}^{e}(\vec{U}) = p_{i,j,k,c,r,t}$.

C.2 Estimation of Selection Probability

To estimate occupational selection probability, we follow Eckardt (2026) and adopt a flexible machine learning approach to avoid functional form assumption of the underlying sub-utility function. We predict occupational choice conditional on each major by using a random forest algorithm. The advantage of the random forest lies in its flexibility in incorporating a large number of predictors into the model and its prediction accuracy. Meanwhile, it avoids the usual drawbacks of the conditional logit model that suffers from the independence of irrelevant alternatives property.

In practice, our explanatory variables for predicting occupational choice include graduation cohort dummies, age, gender, work experience, bands of HELP loan repayments, and a

[37] We assume the correlation differs only across major-by-occupation-set combinations (Section 4.1.2).

set of local labor-demand shifters that serve as exclusion restrictions. Following Kinsler and Pavan (2015), we construct the shifters from the full tax sample for 2003–2019, without the restrictions that define the estimation sample. Let $\pi_{k,r,t}$ denote the share of occupation k in employment in region r and year t, and $\pi_{k,t}$ its national share. Separately for each occupation k, we estimate across regions and years

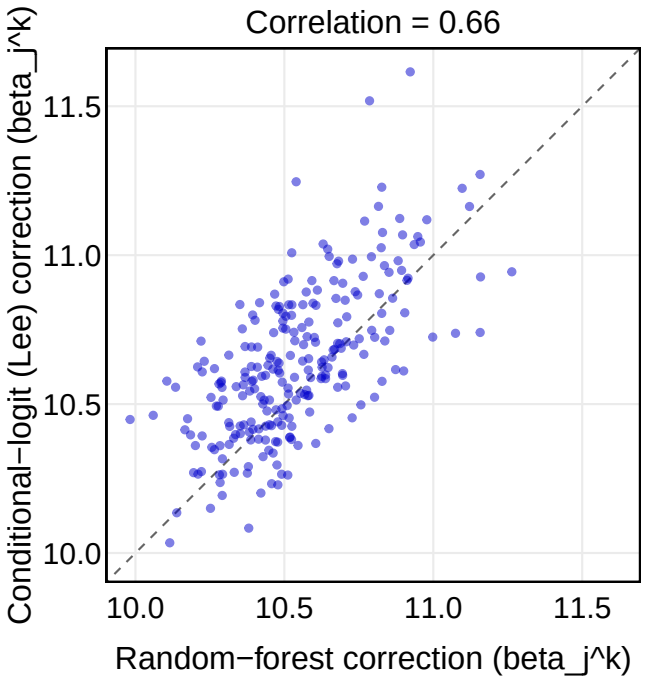
$$\pi_{k,r,t} = a_k + b_k \pi_{k,t} + \bar{\mathbf{x}}'_{k,r,t}\mathbf{c}_k + Z_{k,r,t}, \tag{C.9}$$

where $\bar{\mathbf{x}}_{k,r,t}$ collects the mean real annual earnings, the mean age, and the share of men among workers in occupation k, region r, and year t. The shifter $Z_{k,r,t}$ is the residual, the part of occupation k's local employment share that is accounted for neither by its national share nor by the earnings and demographic composition of the local occupation cell. Because occupational choice is predicted conditional on major, we aggregate the shifters to the occupation sets of Section 4.1.2: for major j and set g, $Z_{j,g,r,t}$ is the employment-weighted mean of $Z_{k,r,t}$ over the occupations k in set g, set to zero where a set has no workers in the region-year. Each individual is assigned the shifters of their major, region, and year, measured contemporaneously.

The identifying assumption is that, conditional on the controls in the earnings equation (20), the shifters move the probability of choosing an occupation set without entering earnings directly. Because Equation (C.9) conditions on the mean earnings of the local occupation cell, $Z_{k,r,t}$ is, by construction, uncorrelated within each occupation with the mean earnings of its region-year cell, so it does not capture local specialization in occupations that pay more in that region and year. This does not rule out persistent local demand shocks that affect individual earnings beyond the cell mean, and we treat exclusion as a maintained assumption.

To check that our selection-corrected returns are not an artifact of this non-parametric first stage, we re-estimate the first stage with a parametric conditional logit and form the corresponding Lee (1983) correction, holding the sample, instruments, and outcome fixed. Figure C1 plots the resulting major-occupation returns $\bar{\beta}^k_j = \frac{1}{T}\sum_t^T \beta^k_{j,t}$ against the random-forest baseline. The mean absolute difference is 0.17 log points, with a correlation of 0.66 across the 244 major-occupation cells, and the top occupation of every major is the same under both corrections. The within-major return differences that drive mismatch do move, however. Computing the micro mismatch of Equation (9) on these bin-averaged returns, with the same efficiency-unit weights and top occupations, gives 15.5 percent under the conditional logit against 21.3 percent under the random forest; the fall is concentrated in Health, Education, and Society and Culture, while Management and Commerce and Creative Arts are little changed. The aggregate counterpart under the conditional logit is not recomputed. We retain the random forest as the baseline, following Eckardt (2026).

Figure C1: Selection-Corrected Returns: Random Forest vs. Conditional Logit



Notes: Each point is a major-occupation cell. The horizontal axis is the selection-corrected return β_j^k using the random-forest first stage; the vertical axis uses a parametric conditional-logit (Lee) correction with the same sample and instruments. The dashed line is the 45° line.

D Estimation of Production Function Parameters

The first order condition of the firm yields,

$$\underbrace{\ln\left(p^k_{j',r,t}/p^k_{j,r,t}\right)}_{\beta^k_{j',r,t}-\beta^k_{j,r,t}} = \ln\left(\alpha^k_{j',t}/\alpha^k_{j,t}\right) + (\sigma_k - 1)\ln\left(H^k_{j',r,t}/H^k_{j,r,t}\right); \quad \text{for each } k \in K, \tag{D.1}$$

where $p^k_{j',r,t}$ is the return to major-occupation combination conditional on the region r and year-bin t; and $H^k_{j',r,t} = \sum_{i \in I^k_{j',r,t}} \tilde{h}_i$ is total efficiency labor units. We obtain the region-specific returns by re-estimating the selection-corrected earnings equation (20) separately in each region (GCCSA) and year bin, with major–occupation dummies, the same individual controls, and control-function coefficients re-estimated in each region-year bin. Each $\beta^k_{j,r,t} = \ln p^k_{j,r,t}$ therefore also contains the earnings level of its region-year bin. We keep major–occupation cells with at least five workers in the region-year bin and occupations with at least two majors present, and compute $H^k_{j,r,t}$ from the same subsample. Because the base major of an occupation is not observed in every region-year bin, we normalize returns and efficiency units relative to a single reference cell per occupation rather than within each region-year bin. For each occupation k, the estimating equation is therefore a regression of $\beta^k_{j,r,t}$ on $\ln H^k_{j,r,t}$ with major fixed effects, and major-specific linear time trends where the weights $\alpha^k_{j,t}$ vary over time. Differences in earnings levels across region-year bins that are common to all majors within an occupation are not differenced out, so to the extent that they are correlated with cell size they affect the estimate of σ_k. Table 5 reports how the aggregate results vary with σ_k. To this end, for each occupation k, we obtain the estimates of σ_k and $\{\alpha^k_{j,t}\}$ with the constraint of $\sum_{j \in J} \alpha^k_{j,t} = 1$. The weights are estimated only for the 244 major–occupation cells in the estimation sample. Cells with fewer than 100 observations lie outside the planner's choice set, which consists of the in-sample cells. Because a larger choice set can only raise the output the planner can attain, measured macro mismatch is conservative with respect to this restriction.

For the aggregate production function, estimating $\{\alpha_{k,t}\}$ and σ from Equation (26) needs variation in $p^{k'}_{r,t}$ and $Y^{k'}_{r,t}$. We continue to exploit cross-regions and year-bin variation. By Equation (5), we obtain $Y^k_{r,t}$ using estimated $p^k_{j',r,t}$ and $H^k_{j',r,t}$. Equation (26) can then be estimated at the level of region and year-bin,

$$\ln\left(\frac{\left(\sum_j \left(\alpha^{k'}_{j,t}\right)^{-\frac{1}{\sigma_{k'}-1}} \left(p^{k'}_{j,r,t}\right)^{\frac{\sigma_{k'}}{\sigma_{k'}-1}}\right)^{(1-\frac{1}{\sigma_{k'}})}}{\left(\sum_j \left(\alpha^k_{j,t}\right)^{-\frac{1}{\sigma_k-1}} \left(p^k_{j,r,t}\right)^{\frac{\sigma_k}{\sigma_k-1}}\right)^{(1-\frac{1}{\sigma_k})}}\right) = \ln\left(\frac{\alpha_{k',t}}{\alpha_{k,t}}\right) + (\sigma - 1)\ln\left(\frac{Y^{k'}_{r,t}}{Y^k_{r,t}}\right). \tag{D.2}$$

As discussed in Section 2, the weight $\alpha_{k,t}$ already absorbs occupation-specific productivity, so the two cannot be told apart in (26); this is not crucial for our exercise. A typical assumption in the literature (see Katz and Murphy, 1992) is to assume a linear time trend in technological change, e.g., the expression can be approximated by,

$$\ln\left(\frac{\alpha_{k',t}}{\alpha_{k,t}}\right) = \gamma_{0,kk'} + \gamma_{1,kk'}t$$

when using time-series data or simply $\gamma_{0,kk'}$ if using only cross-sectional variation. In the technical-change specification, we impose the same restriction on the within-occupation weights, $\ln(\alpha^k_{j',t}/\alpha^k_{j,t}) = \gamma^k_{0,jj'} + \gamma^k_{1,jj'}t$, by adding major-specific linear trends in the year-bin index to the regression for each occupation, and occupation-specific linear trends to the regression for the aggregate production function. Both sets of trends are estimated jointly over all year bins, and the levels of the weights in each year bin follow from the adding-up constraints. In the baseline, the trend terms are omitted, so all weights are constant over time.

To this end, we obtain the estimates σ and $\{\alpha_{k,t}\}$ by the constraint of $\sum_{k\in K}\alpha_{k,t} = 1$. Therefore, all production function parameters $\{\alpha^k_{j,t}\}$, $\{\sigma_k\}$, $\{\alpha_{k,t}\}$, and σ are estimated. Table D1 provides the estimated production function parameters.

Standard errors in parentheses in Table D1 are obtained from the parametric bootstrap described in Appendix D.1.

D.1 Inference

We quantify variance for two sets of objects, the life-cycle mismatch series in Figure 1 and the production-function parameters in Table D1, with a two-tier procedure.

Micro-estimate band (analytic). Mismatch in Equations (8) and (9) is a smooth function $h(\boldsymbol{\beta})$ of the estimated major-occupation returns $\boldsymbol{\beta}$. We obtain its variance by the multivariate delta method, $Var(h(\boldsymbol{\beta})) = \nabla h(\boldsymbol{\beta})^T\, Cov(\boldsymbol{\beta})\, \nabla h(\boldsymbol{\beta})$, using the selection-corrected coefficient covariance. We apply this age by age, reusing the age-stratified efficiency-unit weights from the life-cycle estimation, to produce the 95% band on the micro series in Figure 1. As in Table 3, the individual efficiency units $\tilde{h}_i$ are held fixed.

Production-function parameters and macro-estimate band (parametric bootstrap). We draw $\boldsymbol{\beta}^{(b)} \sim N(\hat{\boldsymbol{\beta}}, \hat{V}_{\boldsymbol{\beta}})$ for $b = 1,\ldots,B$ with $B = 200$, using the selection-corrected covariance. For each draw we re-estimate the occupation- and aggregate-production parameters $(\{\alpha^k_{j,t}\}, \{\sigma_k\}, \{\alpha_{k,t}\}, \sigma)$; the standard errors reported in Table D1 are the standard deviations of these estimates across draws. The same draws are then propagated through the equilibrium solver: for each draw we recompute equilibrium prices and the age-specific mismatch series, and the shaded macro-

Table D1: Estimates of Production Function Parameters

				Major Weights Within Occupation									
ANZSCO 2-digit	Occupation Title	α_k	Elas	1	2	3	4	5	6	7	8	9	10
21	Arts and Media Professionals	0.023	2.120	–	–	–	–	–	0.190	0.168	0.217	0.212	0.213
		(0.001)	(0.000)						(0.017)	(0.018)	(0.012)	(0.012)	(0.008)
22	Business, Human Resource and Marketing Professionals	0.084	1.927	0.075	0.080	0.073	0.057	0.066	0.069	0.055	0.368	0.094	0.062
		(0.001)	(0.000)	(0.003)	(0.003)	(0.003)	(0.006)	(0.005)	(0.002)	(0.002)	(0.004)	(0.002)	(0.001)
23	Design, Engineering, Science and Transport Professionals	0.077	2.254	0.131	0.068	0.260	0.113	0.101	0.084	0.060	0.087	0.072	0.025
		(0.001)	(0.000)	(0.002)	(0.004)	(0.003)	(0.002)	(0.002)	(0.002)	(0.005)	(0.002)	(0.003)	(0.000)
24	Education Professionals	0.047	1.801	0.055	0.039	0.045	–	0.044	0.094	0.529	0.076	0.084	0.033
		(0.001)	(0.000)	(0.004)	(0.005)	(0.004)		(0.006)	(0.002)	(0.006)	(0.002)	(0.003)	(0.001)
25	Health Professionals	0.040	1.660	0.044	–	0.029	–	0.038	0.715	0.030	0.046	0.043	0.054
		(0.001)	(0.000)	(0.003)		(0.003)		(0.007)	(0.009)	(0.003)	(0.003)	(0.003)	(0.005)
26	ICT Professionals	0.056	1.937	0.073	0.268	0.168	–	0.071	0.061	0.061	0.142	0.083	0.073
		(0.001)	(0.000)	(0.004)	(0.006)	(0.005)		(0.012)	(0.006)	(0.007)	(0.004)	(0.006)	(0.004)
27	Legal, Social and Welfare Professionals	0.034	1.810	0.086	–	0.105	–	–	0.082	0.052	0.164	0.492	0.019
		(0.001)	(0.000)	(0.014)		(0.011)			(0.007)	(0.005)	(0.006)	(0.011)	(0.001)
31	Engineering, ICT and Science Technicians	0.044	1.907	0.107	0.116	0.138	0.112	0.071	0.094	0.064	0.117	0.091	0.091
		(0.001)	(0.000)	(0.003)	(0.004)	(0.004)	(0.003)	(0.005)	(0.004)	(0.005)	(0.004)	(0.005)	(0.004)
34	Electrotechnology and Telecommunications Trades Workers	0.035	1.772	0.080	0.142	0.167	0.067	0.078	0.090	0.079	0.141	0.093	0.064
		(0.001)	(0.000)	(0.005)	(0.006)	(0.006)	(0.009)	(0.009)	(0.006)	(0.005)	(0.005)	(0.005)	(0.004)
36	Skilled Animal and Horticultural Workers	0.022	1.693	0.122	–	0.096	0.098	0.125	0.115	0.094	0.119	0.107	0.124
		(0.001)	(0.000)	(0.008)		(0.011)	(0.015)	(0.007)	(0.009)	(0.007)	(0.007)	(0.008)	(0.012)
3X	Automotive, Engineering, and Construction Trades Workers	0.036	1.835	0.076	0.082	0.122	0.108	0.097	0.088	0.098	0.126	0.112	0.090
		(0.001)	(0.000)	(0.007)	(0.007)	(0.005)	(0.008)	(0.009)	(0.005)	(0.007)	(0.004)	(0.006)	(0.005)
3Y	Food Trades, Other Technicians, and Trades Workers	0.028	1.778	0.086	0.086	0.125	0.109	0.099	0.104	0.094	0.145	0.103	0.050
		(0.001)	(0.000)	(0.006)	(0.007)	(0.006)	(0.013)	(0.010)	(0.007)	(0.006)	(0.005)	(0.006)	(0.002)
41	Health and Welfare Support Workers	0.028	1.800	0.077	–	–	–	–	0.274	0.112	0.116	0.213	0.209
		(0.001)	(0.000)	(0.012)					(0.009)	(0.008)	(0.007)	(0.007)	(0.016)
42	Carers and Aides	0.032	2.013	0.085	0.078	0.076	–	0.074	0.142	0.168	0.126	0.153	0.099
		(0.001)	(0.000)	(0.009)	(0.009)	(0.009)		(0.013)	(0.005)	(0.005)	(0.006)	(0.005)	(0.006)
43	Hospitality Workers	0.030	1.958	0.085	0.084	0.092	0.096	0.066	0.113	0.100	0.182	0.118	0.062
		(0.000)	(0.000)	(0.005)	(0.008)	(0.006)	(0.008)	(0.006)	(0.003)	(0.003)	(0.005)	(0.004)	(0.002)
45	Sports and Personal Service Workers	0.026	1.814	0.092	–	0.104	–	0.089	0.144	0.139	0.205	0.165	0.062
		(0.001)	(0.000)	(0.010)		(0.011)		(0.011)	(0.005)	(0.006)	(0.006)	(0.007)	(0.003)
52	Personal Assistants and Secretaries	0.021	1.868	0.112	–	–	–	0.117	0.112	0.126	0.262	0.181	0.090
		(0.001)	(0.000)	(0.012)				(0.014)	(0.008)	(0.007)	(0.010)	(0.009)	(0.006)
53	General Clerical Workers	0.038	1.895	0.077	0.073	0.099	0.084	0.071	0.108	0.125	0.191	0.130	0.040
		(0.000)	(0.000)	(0.005)	(0.005)	(0.005)	(0.007)	(0.005)	(0.003)	(0.004)	(0.004)	(0.003)	(0.002)
54	Inquiry Clerks and Receptionists	0.026	2.005	0.101	0.102	0.117	–	0.080	0.120	0.111	0.199	0.147	0.022
		(0.001)	(0.000)	(0.010)	(0.010)	(0.009)		(0.010)	(0.006)	(0.005)	(0.007)	(0.007)	(0.001)
55	Numerical Clerks	0.048	1.993	0.071	0.090	0.080	0.060	0.080	0.096	0.089	0.243	0.109	0.081
		(0.001)	(0.000)	(0.005)	(0.006)	(0.005)	(0.010)	(0.008)	(0.004)	(0.004)	(0.006)	(0.004)	(0.004)
56	Clerical and Office Support Workers	0.021	1.704	0.126	0.116	0.108	–	–	0.124	0.119	0.185	0.149	0.073
		(0.001)	(0.000)	(0.020)	(0.013)	(0.011)			(0.012)	(0.013)	(0.010)	(0.010)	(0.006)
59	Other Clerical and Administrative Workers	0.038	1.655	0.079	0.080	0.082	0.067	0.091	0.084	0.077	0.202	0.152	0.087
		(0.001)	(0.000)	(0.005)	(0.008)	(0.005)	(0.011)	(0.010)	(0.005)	(0.004)	(0.005)	(0.005)	(0.005)
61	Sales Representatives and Agents	0.045	1.857	0.066	0.079	0.081	0.075	0.096	0.102	0.090	0.269	0.105	0.036
		(0.001)	(0.000)	(0.005)	(0.006)	(0.004)	(0.010)	(0.007)	(0.004)	(0.004)	(0.006)	(0.004)	(0.001)
62	Sales Assistants and Salespersons	0.041	2.211	0.098	0.089	0.090	0.100	0.076	0.099	0.111	0.149	0.129	0.058
		(0.000)	(0.000)	(0.004)	(0.004)	(0.004)	(0.008)	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)	(0.001)
63	Sales Support Workers	0.031	1.754	0.080	0.078	0.090	0.099	0.093	0.099	0.110	0.164	0.107	0.078
		(0.001)	(0.000)	(0.008)	(0.008)	(0.007)	(0.015)	(0.011)	(0.005)	(0.006)	(0.006)	(0.005)	(0.004)
7X	Machinery Operators and Drivers	0.035	1.787	0.095	0.085	0.133	0.084	0.078	0.101	0.098	0.159	0.115	0.051
		(0.001)	(0.000)	(0.007)	(0.006)	(0.006)	(0.009)	(0.007)	(0.005)	(0.004)	(0.004)	(0.005)	(0.002)
8X	Labourers	0.015	2.082	0.093	0.082	0.116	0.099	0.093	0.095	0.089	0.169	0.110	0.053
		(0.000)	(0.000)	(0.004)	(0.004)	(0.004)	(0.007)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
	Elasticity of Substitution of Occupational Products		2.332										
			(0.002)										

Notes: This table shows the estimated production function parameters. Baseline specification: all parameters are constant over time (Appendix D); the trend parameters of the technical-change specification used in Section 5.3 are not reported. Columns 5-14 show the major weights within each occupation, with the numbers 1-10 representing the 2-digit ASCED major. A dash marks major-occupation cells with fewer than 100 observations, which are excluded from the estimation sample; their weights are not estimated and the planner does not allocate workers to them. Standard errors (in parentheses) are obtained from a parametric β-bootstrap of the wage-regression vcov (B = 200 draws), propagated through the production-function estimation; see Appendix D, §Inference. The standard errors of the within-occupation elasticities are zero because a draw shifts each cell's return uniformly across regions and year bins, which the major fixed effects absorb; the bootstrap therefore does not capture uncertainty in these elasticities (see Table 5).

estimate band in Figure 1 is ±1.96 times the standard deviation of that series across draws. The same draws also deliver the standard errors on major-specific macro mismatch in Table 3. The bootstrap redraws only the return coefficients: the first-stage choice probabilities, the individual efficiency units, the regional quantities entering the production-function estimation, and the top occupations are held at their estimated values, so all standard errors and bands are conditional on these generated inputs. Because a draw shifts a cell's return by the same amount in every region and year bin, and the major fixed effects in the elasticity regressions ab-

sorb such a shift, the within-occupation substitution parameters $\{\sigma_k\}$ do not vary across draws; their reported standard errors are zero, and the bootstrap does not capture uncertainty in these elasticities. Table 5 reports the sensitivity of the aggregate results to them instead.

E Robustness

E.1 Mismatch with Occupational Tenures and Variable Prices

E.1.1 Planner's Problem

Consider the tenure-effective aggregate human capital of major j in occupation k, which weights each worker's efficiency units $\tilde{h}_i^k$ by the gross return to occupational tenure $e^{\tilde{\tau}_i}$, where $\tilde{\tau}_i \equiv \hat{\eta}_{1,j,k}\tau_i + \hat{\eta}_{2,j,k}\tau_i^2$ is the fitted log tenure premium from Equation (28),

$$\tilde{H}_j^k = \sum_{i \in I_j^k} \tilde{h}_i^k e^{\tilde{\tau}_i} = \sum_{i \in I_j^k} \tilde{h}_i^k \frac{\sum_{i \in I_j^k} \tilde{h}_i^k e^{\tilde{\tau}_i}}{\sum_{i \in I_j^k} \tilde{h}_i^k} = H_j^k T_j^k, \tag{E.1}$$

where $H_j^k = \sum_{i \in I_j^k} \tilde{h}_i^k$ is the base (tenure-free) human capital and $T_j^k = \frac{\sum_{i \in I_j^k} \tilde{h}_i^k e^{\tilde{\tau}_i}}{\sum_{i \in I_j^k} \tilde{h}_i^k} = E\left[e^{\tilde{\tau}_i} \mid j,k\right]$ is the expected gross return to tenure. In the implementation, T_j^k is the sample mean of $e^{\tilde{\tau}_i}$ within each major–occupation cell and year bin, which coincides with the expression above when tenure premia and efficiency units are uncorrelated within the cell. Each occupation produces with these effective inputs, $Y^k = F^k(\tilde{H}_1^k, \ldots, \tilde{H}_J^k) = F^k(H_1^k T_1^k, \ldots, H_J^k T_J^k)$.

Now suppose the planner reallocates labor across occupations but, holding the within-occupation tenure distribution fixed, cannot choose *which* workers move. Let $\hat{H}_j^k$ denote the post-reallocation base allocation of major j to occupation k, satisfying market clearing $\sum_k \hat{H}_j^k = \bar{H}_j \equiv \sum_k H_j^k$ for every major j. Workers who remain in a cell keep their tenure premium T_j^k, while workers who newly enter, arrive without occupational tenure, so each cell is either a net sender ($\hat{H}_j^k \le H_j^k$) or a net receiver ($\hat{H}_j^k > H_j^k$). The tenure-effective input under the reallocation is therefore the piecewise-linear function

$$\tilde{H}_j^k\left(\hat{H}_j^k\right) = \begin{cases} \hat{H}_j^k T_j^k, & \hat{H}_j^k \le H_j^k \quad \text{(net outflow)}, \\ H_j^k T_j^k + \left(\hat{H}_j^k - H_j^k\right), & \hat{H}_j^k > H_j^k \quad \text{(net inflow)}, \end{cases} \tag{E.2}$$

which reduces to the observed effective allocation (E.1) when $\hat{H}_j^k = H_j^k$. This map defines the *lower bound* of Section 6.1, in which tenure returns are occupation-specific and forfeited on switching. The *upper bound* instead treats tenure returns as transferable, so that entrants are credited with the destination cell's premium and $\tilde{H}_j^k(\hat{H}_j^k) = \hat{H}_j^k T_j^k$ for all $\hat{H}_j^k$. This map is linear, and the planner's problem is then the baseline problem with weights $\alpha_j^k (T_j^k)^{\sigma_k}$ in place of α_j^k. Aggregate output under the reallocation is $Y(\hat{\mathbf{H}}) = F\left(F^1(\tilde{\mathbf{H}}^1), \ldots, F^K(\tilde{\mathbf{H}}^K)\right)$, where

$\tilde{\mathbf{H}}^k = (\tilde{H}_1^k, \ldots, \tilde{H}_J^k)$ and $\hat{\mathbf{H}} = \{\hat{H}_j^k\}$.

The planner chooses the allocation $\hat{\mathbf{H}} = \{\hat{H}_j^k\}$ to maximize aggregate output $Y(\hat{\mathbf{H}})$ subject to the per-major resource constraints,

$$\max_{\hat{\mathbf{H}} \in \Delta} Y(\hat{\mathbf{H}}), \qquad \Delta = \left\{ \hat{\mathbf{H}} : \ \hat{H}_j^k \geq 0, \ \textstyle\sum_k \hat{H}_j^k = \bar{H}_j \ \ \forall j \right\}. \tag{E.3}$$

This is output maximization over a product of J simplexes, one per major, given by the market-clearing constraints (11). The baseline macro estimate is the special case $T_j^k \equiv 1$, where $\tilde{H}_j^k = \hat{H}_j^k$.

E.1.2 Computational Procedure

We solve the planner's program (E.3) by maximizing aggregate output directly, rather than by root-finding the first-order conditions.

Concavity. Our estimates place both substitution parameters in the interval $[0, 1)$. The elasticity between occupations is $1/(1-\sigma) = 2.33$, so $\sigma = 0.571$, and every within-occupation is $\sigma_k \in [0, 2/3]$ (see Table D1 and the sensitivity grid in Table 5). For $\sigma_k \leq 1$ each occupational aggregator F^k is concave in its effective inputs $(\tilde{H}_1^k, \ldots, \tilde{H}_J^k)$, and for $\sigma \leq 1$ the final aggregator F is concave and nondecreasing in $(Y^1, \ldots, Y^K)$; their composition is therefore concave in $\tilde{\mathbf{H}}$. Because the substitution parameters lie below one, the marginal product of a major in an occupation grows without bound as its input vanishes, so the efficient allocation is interior and no cell in the planner's choice set is left inactive. The tenure map (E.2) is continuous, increasing, and concave in $\hat{H}_j^k$ (its slope falls from $T_j^k > 1$ below H_j^k to 1 above), so Y remains concave in $\hat{\mathbf{H}}$ once tenure is included. Since Δ is convex, Equation (E.3) is a concave program with a *unique* global maximizer $\hat{\mathbf{H}}^*$. This is what justifies solving it by ascent rather than by selecting among the multiple roots of the stationarity conditions (10).

Algorithm. For each year group we maximize (E.3) by a mirror-ascent (multiplicative-weights) iteration, which keeps the allocation strictly inside the per-major simplex and respects the resource constraints by construction:

1. *Initialize* at the observed allocation, $\hat{\mathbf{H}}^{(0)} = \mathbf{H}$, which is feasible ($\sum_k H_j^k = \bar{H}_j$).
2. *Marginal products.* At iterate $\hat{\mathbf{H}}^{(s)}$ evaluate the marginal value product of each cell, $p_j^k = \partial Y / \partial \hat{H}_j^k = \left(\partial Y / \partial \tilde{H}_j^k\right) \cdot \left(\partial \tilde{H}_j^k / \partial \hat{H}_j^k\right)$. The first factor is given by the marginal-product in Equation (7) and the second equals T_j^k for net-outflow cells and 1 for net-inflow cells. In the Cobb–Douglas limit $\sigma_k = 0$, $F^k = \prod_j (\tilde{H}_j^k)^{\alpha_j^k}$ and the same gradient applies.

3. *Multiplicative update.* Shift labor toward higher-return occupations within each major and renormalize to the major's fixed supply,

$$\hat{H}_j^{k,(s+1)} = \bar{H}_j \frac{\hat{H}_j^{k,(s)} \exp\left(\eta\, p_j^k\right)}{\sum_{k'} \hat{H}_j^{k',(s)} \exp\left(\eta\, p_j^{k'}\right)},$$

with the step size $\eta > 0$ chosen by a backtracking line search so that aggregate output rises. The update is scale-invariant cell by cell (suited to allocations spanning orders of magnitude) and never leaves the simplex.

4. *Stop* when the marginal products are equalized within each major up to a tolerance. The planner FOC and concavity guarantee a global optimum is reached.

Because each step is a feasible ascent from a feasible start, the procedure improves monotonically and guarantees $Y(\hat{\mathbf{H}}) \geq Y(\mathbf{H})$ at every iterate, so the efficient allocation $\hat{\mathbf{H}}^*$ weakly dominates the observed one and measured mismatch is non-negative. Concavity removes the need for multi-start seeds, and any outer iteration over which cells are net entrants versus leavers with tenure.

At the optimum the marginal products are equalized within each major, recovering the frictionless prices $\hat{p}_j$ of Equation (13); aggregate mismatch then follows from Equation (14) as the proportional output gap between $Y(\hat{\mathbf{H}}^*)$ and $Y(\mathbf{H})$.

E.2 Mismatch with Self-Selection Effects

The baseline mismatch measure values every worker at the population return to their major in each occupation. This appendix asks how mismatch changes when a worker's current wage also reflects how they were selected into their occupation. The control function in Equation (20) identifies the selection term $E[\epsilon_{i,j,k} \mid M_{i,j,k} = 1, W_{i,t}]$ only for workers observed in occupation k; it does not identify the earnings shock that the same worker would draw in an occupation they did not choose. We therefore set the mean of that shock to zero, $E[\epsilon_{i,j,k'} \mid M_{i,j,k'} = 0, W_{i,t}] = 0$, so that in an occupation they did not choose, a worker earns what a worker of the same major and observable characteristics earns on average. Under the selection model of Section 4.1.2, the selection term of a worker who did not choose k' is $-\hat{p}_{i,j,k'}/(1-\hat{p}_{i,j,k'})$ times that of a worker who did, so the assumption is close to the model's own implication for occupations a worker was unlikely to choose.

Under this assumption, the wage of individual i with major j, currently in occupation k, in

any occupation k' is

$$w_{i,j,k'} = p_j^{k'}\,\hat{h}_i^{k'}, \quad \text{where} \quad \hat{h}_i^{k'} \equiv \begin{cases} \tilde{h}_i \cdot \exp\!\left[\hat{\rho}_j^k\, g\!\left(\hat{p}_{i,j,k}\right)\right], & k' = k, \\ \tilde{h}_i, & k' \neq k, \end{cases} \tag{E.4}$$

with $g(\hat{p}_{i,j,k}) = -\phi\!\left(\Phi^{-1}(\hat{p}_{i,j,k})\right)/\hat{p}_{i,j,k}$ the inverse Mills ratio evaluated at the estimated probability that individual i chooses occupation k, and $\hat{\rho}_j^k$ the estimated selection coefficient for the (j,k) cell. In the occupation the worker chose, the selection-inclusive efficiency $\hat{h}_i^k$ combines base productivity $\tilde{h}_i$ with the expected selection effect given the worker's estimated choice probability; the worker's realized shock remains in the residual. When workers are positively selected into occupation k, because unobserved factors that raise wages also raise the probability of choosing k, $\hat{h}_i^k > \tilde{h}_i$, and conversely for negative selection.

The selection-inclusive mismatch measure differs from the baseline measure in Equation (8) in two respects. First, a worker's current wage includes the selection effect, $p_j^k \hat{h}_i^k$, so the denominator reflects what the worker actually earns, including any selection premium or penalty. Second, each worker's destination is their individually optimal occupation,

$$k^*(i) = \arg\max_{k' \in \mathcal{K}_j} p_j^{k'}\,\hat{h}_i^{k'}, \tag{E.5}$$

where $\mathcal{K}_j$ is the set of occupations available to major j. Because $\hat{h}_i^{k'} = \tilde{h}_i$ for every $k' \neq k$, the worker compares the population return in each alternative occupation with their own selection-inclusive wage in their current one. A worker who is strongly positively selected into their current occupation may find that no alternative pays more and stays, $k^*(i) = k$, while a worker with weak or negative selection moves to the highest-return occupation of their major. Major-specific mismatch with selection effects is then

$$\hat{m}_j^{\text{micro}} = \frac{\sum_{i \in I^j} p_j^{k^*(i)}\,\hat{h}_i^{k^*(i)} - \sum_{i \in I^j} p_j^k\,\hat{h}_i^k}{\sum_{i \in I^j} p_j^k\,\hat{h}_i^k}, \tag{E.6}$$

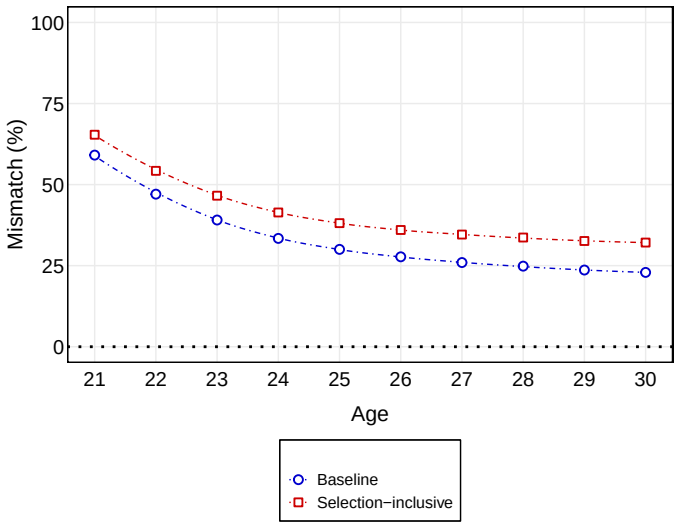
where k is worker i's current occupation, and economy-wide mismatch with selection is

$$\hat{\mathcal{M}}^{\text{micro}} = \frac{\sum_{j \in J}\sum_{i \in I^j} p_j^{k^*(i)}\,\hat{h}_i^{k^*(i)} - \sum_{j \in J}\sum_{i \in I^j} p_j^k\,\hat{h}_i^k}{\sum_{j \in J}\sum_{i \in I^j} p_j^k\,\hat{h}_i^k}. \tag{E.7}$$

This is the wage gain at fixed returns from moving each worker to their individually optimal occupation. It is non-negative by construction, since $k^*(i)$ maximizes $p_j^{k'}\hat{h}_i^{k'}$ and the current

occupation is among the alternatives.

Figure E1: Life-cycle Mismatch with Self-Selection Effects



Notes: The figure plots economy-wide individual mismatch by age for the "Baseline" measure (Equation (9)) and the "Selection-inclusive" measure (Equation (E.7)), which values each worker's current occupation with its selection effect and moves each worker to the individually optimal occupation $k^*(i)$ of Equation (E.5).

Table E1 reports the measure by major and economy-wide, next to the baseline of Table 3 computed on the same person-years. Economy-wide, selection-inclusive mismatch is 35.8 percent, against a baseline of 27.1 percent. The increase does not come from the selection term itself: the average selection term at the current occupation is small and negative (−0.06 log points), and valuing current wages with it while sending every worker to the major's top occupation gives 24.4 percent, close to the baseline (last column). It comes from the destination. Each worker now moves to the individually best-paying occupation of the major, $k^*(i)$, rather than to the top occupation $k^*(j)$, and the gain is largest for Architecture and Building (93.5 percent) and Agriculture (86.7 percent), whose highest-return cells employ only 2 to 5 percent of the major's graduates; the top-occupation rule of Appendix E.3 is designed to avoid such cells. Only 3 percent of the workers outside their major's top occupation have a selection-inclusive current wage above the top occupation's return, so few stay. Figure E1 shows the life-cycle profile: selection-inclusive mismatch is 65 percent at age 21 and 31 percent at age 35, against 59 and 21 percent for the baseline, so the two profiles decline in parallel, with a gap of 8 to 10 percentage points from age 25 onward.

E.3 Defining the Top Occupation

Our mismatch measure compares each worker's return with the return in the "top" occupation for their major. Two design choices enter that benchmark: (i) the share of within-major em-

Table E1: Micro Mismatch with Self-Selection Effects, by Major

	Major	Baseline	Selection-inclusive, $k^*(i)$	Selection-inclusive, top occupation
1	Natural and Physical Sciences	0.219	0.385	0.288
2	Information Technology	0.299	0.442	0.256
3	Engineering and Related Technologies	0.302	0.317	0.251
4	Architecture and Building	0.052	0.935	0.179
5	Agriculture, Environmental and Related Studies	0.147	0.867	0.190
6	Health	0.054	0.148	0.054
7	Education	0.118	0.228	0.087
8	Management and Commerce	0.441	0.451	0.358
9	Society and Culture	0.390	0.423	0.401
10	Creative Arts	0.380	0.313	0.313
	Economy-wide	0.271	0.358	0.244

Notes: Micro mismatch pooled over 2003–2019, as a share of the observed wage bill. "Baseline" is Equation (8) (Table 3, column (4)) computed on the person-years used here. "Selection-inclusive, $k^*(i)$" is Equation (E.6): each worker's current occupation is valued with its selection term and the worker moves to the individually optimal occupation of Equation (E.5). "Selection-inclusive, top occupation" restricts the destination to the major's top occupation $k^*(j)$, with the worker staying when the current selection-inclusive wage is higher.

ployment used to define the top set, and whether (ii) the major-occupation returns β_j^k are the selection-corrected series or the raw conditional mean. Table E2 reports individual mismatch across different specifications of assumption (i). Mismatch is essentially flat across cutoffs of 5%, 10%, and 15% at about 27% with the selection-corrected returns and about 13% with the raw conditional mean. At the 15% cutoff no occupation qualifies for Creative Arts, whose largest occupation employs 13 percent of its graduates, so that column excludes the major; on the same nine majors the 10% cutoff gives 26.3%, so the comparison across cutoffs is flat like for like. Thus, our results are insensitive to the cutoff. The level does depend on the return measure, with correcting for selection roughly doubling the measured mismatch under all sensitivity analyses, consistent with the base estimates in Section 5.1.

Two further definitions, computed from the selection-corrected returns and the same efficiency-unit weights, bracket the baseline. Defining the top occupation as the one employing the largest share of the major's graduates changes the top occupation only for Architecture and Building and for Creative Arts and gives 25.9%. Dropping the share requirement and taking the highest return over all occupations changes the top occupation for six of the ten majors and gives 34.0%; the increase comes from Architecture and Building and from Agriculture, whose highest-return occupations employ 2 to 5 percent of their graduates and whose mismatch rises to 70 and 80 percent. The 10 percent rule is designed to exclude such comparisons with a small, highly paid cell.

Table E2: Economy-wide micro mismatch under alternative top-occupation definitions (%)

β_j^k source	Cutoff 5%	Cutoff 10%	Cutoff 15%
Raw conditional mean	13.1	12.8	12.0
Selection-corrected	27.2	27.1	26.2

Notes:
Each cell reports economy-wide micro mismatch (in %) for a top-occupation cutoff (the share of within-major employment defining the top set) and a source of the major-occupation return β_j^k: the selection-corrected series (baseline) or the raw conditional mean. The baseline cell is (Selection-corrected, 10%). At the 15% cutoff no occupation qualifies for Creative Arts, which is excluded from that column; on the same nine majors the 10% cutoff gives 26.3% with the selection-corrected returns.